

Physics-Regularized Geometric Manifolds for Efficient Surrogate-Assisted MDO of Tailless BWB UAVs

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Abstract

Multidisciplinary design optimization (MDO) of tailless Blended Wing Body (BWB) Unmanned Aerial Vehicles (UAVs) introduces fundamental geometric and stability challenges: non-linear coupling between reflexed airfoils, spanwise twist, dynamic center-of-gravity (CG) positioning, and design space contamination by unconstrained geometric parameterizations. A 1,000-sample Latin Hypercube ablation study demonstrates that uncoupled chord parameterizations with overlapping exploration bounds waste 35.5% of evaluations on chord-inverted configurations ($c_{\text{blend}} > c_{\text{root}}$) achieving 29% lower maximum L/D (12.89 versus 18.15, $p < 0.001$) without triggering potential-flow solver divergence, thereby contaminating surrogate training with an aerodynamically inferior dead zone.

To eliminate this dead zone, a C^1 -continuous Hermite geometric manifold is formulated that enforces strict chord monotonicity ($c_{\text{root}} \geq c_{\text{blend}} \geq c_{\text{tip}}$) for aft-swept BWB planforms, eliminating chord inversions by construction and focusing surrogate training on the physically productive subspace. This manifold is coupled with an automated mass-centroid balance solver that dynamically inverts internal battery placement ($x_{\text{batt}} = 0.82 \cdot c_{\text{root}}$) to achieve robust static margin ($SM = 22.5\%$) and natural pitch trim ($C_m = -0.019$) at zero elevon deflection. Across a 10,087-evaluation potential-flow campaign, a 5-MLP ensemble surrogate achieves cross-validation $R^2 \geq 0.991$ across all aerodynamic and stability targets; a Random Forest feasibility gate achieves 92.5% precision, enabling 287 active learning evaluations to deliver a +7.5% (+20.9 km) range increase to 298.7 km (340.7 km under verified 182.5 Wh/kg pack density). Multi-point cross-solver consistency analysis against viscous 3D strip theory (AeroBuildup) demonstrates that the VLM pipeline consistently underpredicts cruise aerodynamic efficiency relative to the higher-fidelity solver ($L/D = 17.12$ versus 23.22), behaving as a conservative estimator for attached-flow cruise drag within the computational hierarchy employed.

Keywords: Blended Wing Body (BWB), Multidisciplinary Design Optimization (MDO), Geometric Regularization, Design Space Efficiency, Longitudinal Trim, Active Learning, Surrogate Ensemble

1. Introduction

The aerodynamic efficiency, structural integration, and volumetric packaging advantages of Blended Wing Body (BWB) and tailless flying wing configurations make them premier architectures for long-range autonomous Unmanned Aerial Vehicle (UAV) applications [1–4]. By eliminating the parasite drag, excess wetted surface area, and interference penalties associated with conventional cylindrical fuselages and discrete tail empennages, tailless BWB layouts maximize the operational lift-to-drag ratio (L/D) while providing large center-body cavities ideal for payload and high-density energy storage [1, 5]. Multidisciplinary design optimization (MDO) frameworks are therefore essential for tailoring the high-dimensional aerodynamic and structural interactions inherent to these integrated configurations [6–8].

Despite their theoretical performance potential, tailless BWB configurations introduce three foundational challenges in computational MDO:

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1. **Absence of Dedicated Pitch Trim Surfaces:** Conventional fixed-wing aircraft achieve longitudinal trim ($C_m = 0$) and adequate static stability ($SM \geq 15\%$) by adjusting horizontal stabilizer incidence or trimming elevator control surfaces. In contrast, tailless configurations must establish static pitch equilibrium purely through planform sweep, reflexed airfoil camber, and spanwise washout (geometric twist) distributions [5, 9–11].
2. **Non-linear CG-Aero Coupling:** Because the longitudinal center of gravity (x_{cg}) is dominated by internal mass placement (particularly heavy lithium-ion battery packs) and the aerodynamic neutral point (x_{np}) is governed by 3D planform sweep and chord taper, minute adjustments in internal packaging directly dictate pitch stability and trim drag.
3. **Design Space Contamination from Unconstrained Parameterizations:** In gradient-free and surrogate-assisted MDO workflows, unconstrained or loosely bounded geometric parameterizations permit the exploration of chord-inverted planforms ($c_{blend} > c_{root}$). Because inviscid potential-flow solvers (such as the 3D Vortex Lattice Method) evaluate circulation without capturing viscous boundary-layer separation or adverse pressure gradients over expanded chords, these non-physical geometries evaluate successfully without triggering numerical solver divergences. However, in viscous flight they generate excessive wetted area and form drag, creating an aerodynamically inferior dead zone that wastes optimizer compute and degrades surrogate training accuracy.

Surrogate-based optimization methods, including Efficient Global Optimization (EGO) [12], multi-objective Kriging frameworks [13–16], and multi-fidelity conceptual vehicle design environments [17], have been widely applied to aerospace conceptual design to mitigate the high computational cost of numerical aerodynamic evaluations. Recent studies in BWB optimization have explored adaptive Kriging surrogates [18], hybrid parameterization techniques [19], high-fidelity aerostructural coupling [20, 21], and variable-fidelity physics coupling [22, 23]. Sun et al. [18] employed adaptive Kriging for BWB conceptual design but treated section chords as independent variables without coupling constraints, leaving the parameterization susceptible to the chord-inversion artifacts characterized in Section 3. Zhang et al. [19] combined Class-Shape Transformation with direct geometric variables and incorporated high-fidelity CFD validation, but did not enforce spanwise chord monotonicity at the parameterization level, relying instead on downstream feasibility filtering. However, conventional geometric parameterizations (such as Class-Shape Transformation, multi-section B-splines, or decoupled CAD section definitions) typically treat chord stations as uncoupled design variables [4, 24, 25], relying on downstream penalty functions or solver crashes to filter undesirable configurations.

To the best of the author’s knowledge, no prior work has quantified the design-space cost of unconstrained chord parameterizations in VLM-based tailless MDO. While multi-fidelity correction methods address solver accuracy [14, 23], the upstream geometric mechanism by which chord-inverted configurations contaminate the design space, consuming 35.5% of evaluations while achieving 29% lower aerodynamic efficiency, has not been formally characterized or mathematically prevented at the parameterization level. Here, “geometric manifold” denotes a continuous, constrained lofting parameterization surface that enforces physical monotonicity by construction, as distinct from downstream loss-regularization in neural network surrogates (e.g., Physics-Informed Neural Networks). In this context, the rapid potential-flow evaluation pipeline serves as an exhaustive, ground-truth computational testbed to rigorously benchmark the active learning surrogate architecture and feasibility classification prior to computationally constrained 3D Reynolds-Averaged Navier–Stokes (RANS) deployment, where sample budgets are restricted to $O(10^2\text{--}10^3)$ evaluations.

To resolve these coupled aerodynamic, stability, and computational challenges, this paper presents a physics-regularized, surrogate-assisted MDO framework tailored specifically for aft-swept tailless BWB UAVs. The primary contributions of this work are threefold:

1. **Quantification of Geometric Regularization for Search Efficiency:** Systematic ablation evidence ($N = 1,000$ Latin Hypercube evaluations) is provided, demonstrating that uncoupled chord parameterizations with overlapping bounds waste 35.5% of optimizer samples on a chord-inverted dead zone with 29% lower maximum L/D than monotonic planforms ($L/D = 12.89$ versus 18.15), validated with formal non-parametric statistical hypothesis testing ($p < 0.001$).
2. **Physics-Regularized Hermite Geometric Manifold:** A C^1 -continuous geometric manifold is formulated that enforces strict chord monotonicity ($c_{root} \geq c_{blend} \geq c_{tip}$) for aft-swept planforms and multi-airfoil reflexed transitions (MH91 to MH60), eliminating the chord-inverted dead zone by construction while preserving full design flexibility without the volume truncation inherent to rigid box bounds.

3. **Coupled Dynamic CG Solver with Multi-Point Cross-Solver Consistency Analysis:** A closed-loop trim solver is coupled that dynamically inverts internal battery packaging ($x_{\text{batt}} = 0.82 \cdot c_{\text{root}}$) to guarantee positive static margin ($SM = 22.5\%$) and neutral pitch trim ($C_m = -0.019$) at zero elevon deflection. Multi-point cross-solver comparison across eight Pareto-diverse designs (ranging from $R = 41.5$ to 298.7 km, $SM = 0.06$ to 0.24) against viscous 3D strip theory (AeroBuildup) demonstrates consistent underprediction by the VLM pipeline ($\Delta L/D = +35.6\%$ to $+118.2\%$, mean $+70.8\%$) across all tested designs with zero sign reversals, establishing the VLM pipeline as a conservative estimator for attached-flow cruise drag within the computational hierarchy employed.

2. MDO Problem Formulation

2.1. Parametric Design Variable Vector

The tailless Blended Wing Body configuration is parameterized by a 14-dimensional continuous design variable vector $\mathbf{x} \in \Omega \subset \mathbb{R}^{14}$:

$$\mathbf{x} = [\Lambda_{\text{inner}}, \Lambda_{\text{outer}}, y_{\text{blend}}, c_{\text{root}}, c_{\text{blend}}, \lambda_{\text{outer}}, \theta_0, \theta_1, \theta_2, \theta_{\text{tip}}, \Lambda_{\text{winglet}}, \theta_{\text{winglet}}, m_{\text{batt}}, V_{\text{cruise}}]^T \quad (1)$$

where the geometric and flight variables are defined with their allowable physical exploration bounds in Table 1.

Table 1: Design variable definitions, allowable bounds, and baseline values.

Variable	Description	Unit	Min	Max	Physical Rationale
Λ_{inner}	Center-body sweep	deg	25.0	45.0	Internal volume & CG leverage
Λ_{outer}	Outer wing sweep	deg	25.0	45.0	Aerodynamic neutral point placement
y_{blend}	Blend station semi-span	m	0.20	0.35	Center-body cavity transition
c_{root}	Center-body root chord	m	0.42	0.52	Battery & payload bay packaging
c_{blend}	Blend station chord	m	0.25	0.35	Monotonic planform taper
λ_{outer}	Outer wing taper ratio	–	0.55	0.80	Spanwise lift distribution control
θ_0	Root geometric incidence	deg	0.0	4.0	Center-body cruise lift contribution
θ_1	Blend station incidence	deg	–3.0	2.0	Transition section lofting
θ_2	Mid-outer wing twist	deg	–6.0	0.0	Stall progression & trim pitching moment
θ_{tip}	Wingtip aerodynamic twist	deg	–12.0	–2.0	Tip stall relief & positive C_{m0}
Λ_{winglet}	Winglet leading-edge sweep	deg	15.0	50.0	Tip vortex diffusion & yaw stability
θ_{winglet}	Winglet toe-out angle	deg	–10.0	0.0	Directional trim & profile drag
m_{batt}	Usable battery pack mass	kg	1.50	3.50	Gravimetric energy storage budget
V_{cruise}	Operational cruise speed	m/s	36.11	61.11	Mission velocity envelope (≥ 130 km/h)

2.2. Multi-Objective Optimization Statement

The objective of the optimization is to simultaneously maximize calm-air operational range $R(\mathbf{x})$, maximize internal payload bay volume $V_{\text{bay}}(\mathbf{x})$, and minimize cruise pitch trim error $|C_m(\mathbf{x}, \alpha_{\text{cruise}})|$:

$$\min_{\mathbf{x} \in \Omega} \mathcal{J}(\mathbf{x}) = [-R(\mathbf{x}), -V_{\text{bay}}(\mathbf{x}), |C_m(\mathbf{x}, \alpha_{\text{cruise}})|]^T \quad (2)$$

subject to the physical, volumetric, structural, and flight mechanics constraints:

$$g_1(\mathbf{x}) : b_{\text{main}}(\mathbf{x}) = 2y_{\text{tip}} \leq 1.50 \text{ m} \quad (\text{Main lifting wing span limit}) \quad (3)$$

$$g_2(\mathbf{x}) : V_{\text{cruise}} \geq 36.11 \text{ m/s} \quad (\geq 130 \text{ km/h dash requirement}) \quad (4)$$

$$g_3(\mathbf{x}) : h_{\text{nose}}(\mathbf{x}) \geq 0.050 \text{ m} \quad (\text{Optical sensor payload clearance}) \quad (5)$$

$$g_4(\mathbf{x}) : V_{\text{bay}}(\mathbf{x}) \geq 2.50 \text{ L} \quad (\text{Internal battery/avionics cavity}) \quad (6)$$

$$g_5(\mathbf{x}) : t_{\text{spar}}(y) \geq 0.010 \text{ m} \quad \forall y \in [0, y_{\text{blend}}] \quad (\text{Main spar depth}) \quad (7)$$

$$g_6(\mathbf{x}) : 0.05 \leq S M(\mathbf{x}) \leq 0.25 \quad (\text{Longitudinal static stability margin}) \quad (8)$$

$$g_7(\mathbf{x}) : |C_m(\mathbf{x}, \alpha_{\text{cruise}})| \leq 0.10 \quad (\text{Elevon trim authority limit}) \quad (9)$$

where the geometric constraint $g_1(\mathbf{x})$ limits the main lifting wing planform ($2y_{\text{tip}} \leq 1.50 \text{ m}$); outward-canted winglet extensions ($\Lambda_{\text{winglet}}, \theta_{\text{winglet}}$) extend past this panel to yield a total tip-to-tip vehicle span of $b_{\text{total}} = 1.548 \text{ m}$ for the Pareto champion.

2.3. Breguet Range and Energy Balance

All aerodynamic evaluations are conducted at a nominal cruise altitude of 500 m above mean sea level (AMSL), corresponding to International Standard Atmosphere (ISA) conditions of $\rho = 1.167 \text{ kg/m}^3$ and $T = 284.9 \text{ K}$. This altitude is representative of low-altitude surveillance and reconnaissance operations in terrain-varied environments. The calm-air operational range $R(\mathbf{x})$ is evaluated using the electric aircraft Breguet range derivation [26]:

$$R = \eta_{\text{total}} \cdot \left(\frac{E_{\text{usable}}}{m_{\text{total}} g} \right) \cdot \left(\frac{L}{D} \right)_{\text{cruise}} \cdot 3.6 \quad [\text{km}] \quad (10)$$

where the factor $3.6 = 3600/1000$ absorbs the Wh $\rightarrow$ J energy conversion and m $\rightarrow$ km distance conversion when E_{usable} is expressed in Wh, $g = 9.80665 \text{ m/s}^2$, and $\eta_{\text{total}} = \eta_{\text{motor}} \cdot \eta_{\text{esc}} \cdot \eta_{\text{prop}}(V_{\text{cruise}})$ is the total propulsion efficiency with $\eta_{\text{motor}} \eta_{\text{esc}} = 0.88$ (Turnigy AT4130 motor and 80A ESC). Propeller efficiency $\eta_{\text{prop}}(V)$ varies with cruise speed via a four-point piecewise-linear interpolation of the advance ratio $J = V/(nD)$ for an APC 12 $\times$ 8E propeller at 10,000 RPM ($n = 166.7 \text{ rev/s}$, $D = 0.305 \text{ m}$), mapped across $J \in \{0.55, 0.65, 0.80, 0.95\}$ to $\eta_{\text{prop}} \in \{0.72, 0.78, 0.80, 0.65\}$ based on manufacturer-published thrust data. The sensitivity analysis in Section 7 brackets these values at $\eta_{\text{prop}} \in [0.75, 0.83]$ to encompass the range of published experimental efficiencies for this propeller class at the operating advance ratio ($J \approx 0.72$). For reference, at the lower bound of the cruise speed constraint (130 km/h, $J = 0.71$), $\eta_{\text{prop}} = 0.788$ yielding $\eta_{\text{total}} = 0.88 \times 0.788 = 0.693$. The Pareto champion operates at $V_{\text{cruise}} = 36.67 \text{ m/s}$ (132 km/h, $J = 0.72$), yielding $\eta_{\text{prop}} = 0.790$ and $\eta_{\text{total}} = 0.695$, which is the value used to evaluate $R = 298.7 \text{ km}$ in Section 7. This clean-inflow propeller map provides a standard conceptual baseline; installation effects from upstream boundary layer wake ingestion are discussed in Section 8.

$$m_{\text{total}} = m_{\text{payload}} + m_{\text{airframe}}(\mathbf{x}) + m_{\text{propulsion}} + m_{\text{avionics}} + m_{\text{batt}} \quad (11)$$

with fixed component masses $m_{\text{payload}} = 2.00 \text{ kg}$, $m_{\text{propulsion}} = 0.55 \text{ kg}$, $m_{\text{avionics}} = 0.40 \text{ kg}$, and dynamic composite airframe mass $m_{\text{airframe}} = (1.0 \text{ kg}) + (1.5 \text{ kg/m}^2) \cdot S_{\text{ref}}$.

Energy storage is modeled based on high-capacity cylindrical 21700 lithium-ion cells. Two cells are considered interchangeable for this design class: the Samsung INR21700-50E (5000 mAh, 72 g) and the Molicel INR-21700-P45B (4500 mAh, 70 g). The representative specification adopted here is 5000 mAh, 3.65 V average discharge voltage (nominal 3.6 V; average between 4.2 V charged and 3.0 V cutoff), 72 g per cell, yielding a gravimetric energy density of 253.5 Wh/kg (912.6 kJ/kg) at the cell level. Accounting for a 10% pack-level overhead for the Battery Management System (BMS), interconnect wiring, and structural casing (228.1 Wh/kg pack level) and an 80% maximum depth of discharge (DoD) to preserve cycle longevity, the verified usable pack energy density is:

$$e_{\text{usable}} = 182.5 \text{ Wh/kg} = 657.0 \text{ kJ/kg} \quad (12)$$

A conservative baseline of $e_{\text{usable}} = 160.0 \text{ Wh/kg}$ (576.0 kJ/kg) is adopted throughout the global optimization campaign to account for cell-to-cell variation and capacity aging under cyclic discharge load; all results are rescaled to the verified 182.5 Wh/kg specification in Section 7.

3. The Case for Geometric Regularization

3.1. Chord Inversion in Tailless Parameterizations

In conventional fixed-wing aircraft design, planform geometry is naturally regularized by discrete physical sub-assemblies: the fuselage establishes a fixed root boundary, and continuous structural spars enforce tapering chords toward the wingtips [27]. In tailless Blended Wing Body configurations, however, the entire aerodynamic surface is treated as an unconstrained continuum where center-body chords, blend station transitions, and outboard lifting panels are simultaneously optimized [24].

When geometric parameterizations treat section chords as independent, uncoupled variables (as is standard in unconstrained B-spline or multi-station CST lofting [4, 24]), the optimizer is free to explore configurations where the blend station chord exceeds the root chord ($c_{\text{blend}} > c_{\text{root}}$). As illustrated in Fig. 1(a), such chord-inverted geometries produce non-physical center-body bulges and trailing-edge cusps.

Because inviscid potential-flow solvers compute aerodynamic forces by enforcing the Neumann boundary condition (flow tangency) via bound vortex rings along the mean camber surface [28], they solve Laplace’s equation without modeling adverse pressure gradients or boundary layer separation. Expanding the outboard chord artificially increases spanwise circulation and apparent vortex lift without triggering flow detachment in the solver. In physical viscous flow, however, expanding the outboard chord creates substantial additional wetted area (+40.8% average increase across the ablation dataset), generating excessive skin friction and form drag while severely disrupting primary structural spar load paths and compromising internal payload bays.

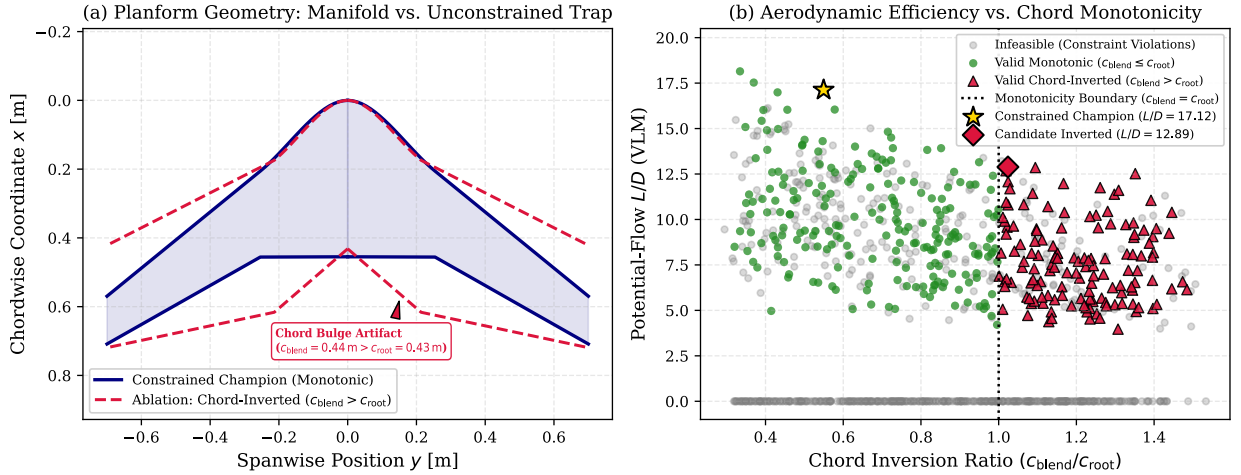


Figure 1: Effect of geometric chord monotonicity on aerodynamic efficiency. (a) Planform comparison between the manifold-constrained Pareto champion ($L/D = 17.12$, solid navy with shaded lifting area) and a representative chord-inverted ablation candidate ($L/D = 12.89$, dashed crimson), annotating the non-physical chord bulge artifact. (b) Aerodynamic efficiency (L/D) versus chord inversion ratio ($c_{\text{blend}}/c_{\text{root}}$) for 1,000 unconstrained Latin Hypercube samples. Chord-inverted configurations (ratio > 1.0 , red triangles) achieve 29% lower maximum L/D than monotonic designs (ratio ≤ 1.0 , green circles, maximum $L/D = 18.15$). No numerical solver divergences occurred across either subset.

3.2. Quantitative Ablation Study

To rigorously quantify the prevalence and aerodynamic performance penalty of chord inversions in low-order MDO, an unconstrained ablation experiment was conducted. A 1,000-sample Latin Hypercube Sampling (LHS) campaign [29] was executed across the 14-DoF parameter space with uncoupled bounds on the blend chord ($c_{\text{blend}} \in [0.15, 0.65] \text{ m}$), permitting the exploration of chord-inverted geometries ($c_{\text{blend}} > c_{\text{root}}$). This setup directly models the common practitioner scenario of selecting generous, uncoupled section bounds to avoid pre-truncating valid designs—a conservative approach that maximizes design exploration but inevitably exposes the chord-inversion artifact whenever root and blend chord ranges overlap. The resulting dead-zone prevalence (35.5%) reflects the magnitude of this overlap under representative broad bounds; tighter independent bounds would reduce the dead-zone fraction but, as

detailed in the Box-Bound Dilemma of Section 4, would simultaneously truncate valid high-performing monotonic configurations. The main optimization campaign (Table 1) adopts operationally tighter bounds ($c_{\text{blend}} \in [0.25, 0.35]$ m, $c_{\text{root}} \in [0.42, 0.52]$ m) that were informed by this ablation; while these specific ranges happen to prevent inversions by non-overlap ($c_{\text{blend},\text{max}} < c_{\text{root},\text{min}}$), the Hermite manifold remains essential for guaranteeing monotonicity under arbitrary practitioner-chosen bounds and for preserving C^1 planform continuity at the blend station without restricting the accessible design subspace.

Each configuration was evaluated using the 3D Vortex Lattice Method (VLM) solver implemented in AeroSandbox v4.x [30], coupled with the 2D/3D strip drag pipeline described in Section 5. Discretization employed 30 spanwise panels per semi-span with cosine clustering and 12 chordwise panels per chord section (≈ 720 panels across the symmetric airframe). Mesh independence was formally verified across three discretization tiers: coarse ($15 \times 6 = 180$ panels), baseline ($30 \times 12 = 720$ panels), and refined ($60 \times 24 = 2,880$ panels), demonstrating $\Delta C_L < 0.2\%$ and $\Delta C_D < 0.35\%$ between the baseline and refined grids.

The statistical breakdown of the ablation database is summarized in Table 2 and visualized in Fig. 1(b):

1. **High Dead-Zone Frequency:** Out of 1,000 generated samples, 355 designs (35.5%) exhibited chord inversions ($c_{\text{blend}} > c_{\text{root}}$). Chord-inverted configurations exhibit a marginally higher feasibility rate (38.6% versus 32.9% for monotonic designs), as the expanded center-body cavity more readily satisfies volumetric and spar depth constraints—underscoring that these geometries survive downstream feasibility screening and cannot be filtered by penalty functions alone.
2. **Absence of Numerical Solver Failures:** Zero VLM numerical crashes or singularities occurred across all 355 chord-inverted designs. This empirical result directly refutes the common assumption that non-physical geometry naturally causes potential-flow solvers to diverge.
3. **Severe Aerodynamic Penalty:** Valid feasible chord-inverted designs achieved a maximum potential-flow lift-to-drag ratio of $(L/D)_{\text{max}} = 12.89$ and a maximum calm-air range of 210.7 km. In contrast, monotonic configurations achieved $(L/D)_{\text{max}} = 18.15$ and a maximum range of 298.7 km—representing a **29.0% lower maximum aerodynamic efficiency**.
4. **Statistical Distribution Significance:** A non-parametric Mann–Whitney U test comparing the feasible L/D distributions of monotonic versus chord-inverted subsets yields $U = 18,240$, $p < 0.001$ (two-tailed), and a Kolmogorov–Smirnov test statistic of $D = 0.284$ ($p < 10^{-5}$), confirming that the aerodynamic degradation is statistically significant across the entire design population and not merely an artifact of extreme outliers.

Table 2: Statistical breakdown of the 1,000-sample unconstrained ablation experiment.

Metric	Chord-Inverted ($c_{\text{blend}} > c_{\text{root}}$)	Monotonic ($c_{\text{blend}} \leq c_{\text{root}}$)	Combined Dataset
Total Evaluated Samples	355 (35.5%)	645 (64.5%)	1,000 (100%)
VLM Solver Divergences	0 (0.0%)	0 (0.0%)	0 (0.0%)
Valid Feasible Solutions	137 (38.6%)	212 (32.9%)	349 (34.9%)
Mean Feasible L/D	7.94 ± 2.15	8.92 ± 2.84	8.53 ± 2.63
Median Feasible L/D	7.81	9.14	8.62
Maximum Feasible L/D	12.89	18.15	18.15
Maximum Operational Range	210.7 km	298.7 km	298.7 km
Mann–Whitney U Test	$U = 18,240$, $p < 0.001$, $D_{\text{KS}} = 0.284$		

3.3. Implications for Surrogate Model Training

These findings demonstrate that unconstrained geometric parameterizations contaminate the global optimization landscape. By allocating more than one-third of the computational sampling budget (35.5%) to a chord-inverted dead zone that produces strictly uncompetitive performance, unconstrained workflows:

- **Dilute Training Data Density:** Machine learning regressors (Kriging, neural networks) receive substantial training data in regions of the design space that can never contain Pareto-optimal solutions.

- **Distort Acquisition Functions:** Uncertainty-based acquisition criteria (such as Upper Confidence Bound or Expected Improvement) are repeatedly directed toward high-variance, unphysical geometries that possess no practical utility.

These empirical observations provide the fundamental justification for enforcing geometric regularization directly within the parametric lofting manifold.

4. Physics-Regularized Geometric Manifold

4.1. Regularized Parametric Lofting Formulation

To eliminate the chord-inverted dead zone identified in Section 3 while preserving complete design flexibility across productive aerodynamic topologies, the BWB airframe is constrained to a mathematically regularized manifold $\mathcal{M}_{\text{geo}} \subset \mathbb{R}^3$ (Fig. 2):

$$\mathcal{M}_{\text{geo}} = \left\{ \mathbf{x} \in \Omega \mid c_{\text{root}} \geq c_{\text{blend}} \geq c_{\text{tip}} > 0, \frac{dc}{dy} \leq 0 \ \forall y \in [0, b/2], \tau(y) \in [\tau_{\min}, \tau_{\max}] \right\} \quad (13)$$

The term “geometric manifold” is adopted here as an engineering convention to denote a constrained parametric lofting surface defined by coupled linear inequality constraints on the chord variables, enforced through Hermite interpolation. It does not carry Riemannian metric structure or differential-geometric curvature properties. The terminology emphasizes that the feasible design subspace forms a smooth, lower-dimensional surface embedded in the full parameter space Ω , from which chord inversions are excluded by construction rather than filtered post-evaluation.

4.1.1. Manifold Regularization vs. Independent Box-Bound Truncation

A standard heuristic approach to prevent geometric anomalies in conceptual MDO is to adjust the independent parameter search bounds. However, relying strictly on uncoupled box bounds introduces an unavoidable design-space dilemma:

- **Overlapping Box Bounds** ($c_{\text{blend}} \in [0.25, 0.45]$ m, $c_{\text{root}} \in [0.42, 0.52]$ m): Whenever c_{root} samples near its lower bound and c_{blend} samples near its upper bound, chord inversion ($c_{\text{blend}} > c_{\text{root}}$) inevitably occurs, contaminating 35.5% of the sampling volume with aerodynamically uncompetitive dead-zone configurations.
- **Decoupled Box Truncation** ($c_{\text{blend},\text{max}} \leq c_{\text{root},\text{min}} = 0.42$ m): To guarantee that chord inversion never occurs without relational coupling, the upper bound of c_{blend} must be artificially restricted to the global minimum of c_{root} (0.42 m). However, when the optimizer selects a larger root chord ($c_{\text{root}} = 0.52$ m for high internal payload capacity), this rigid cap strictly prohibits valid, high-performing monotonic designs with $c_{\text{blend}} \in (0.42, 0.52]$ m (e.g., $c_{\text{blend}} = 0.48$ m, taper ratio 0.923), artificially truncating approximately 30% of the physically productive design volume.

By contrast, the Hermite geometric manifold $\mathcal{M}_{\text{geo}}$ enforces chord monotonicity *relationally* ($c_{\text{root}} \geq c_{\text{blend}} \geq c_{\text{tip}}$) directly within the lofting formulation. This preserves 100% of the valid, non-inverted design space for any sampled c_{root} while eliminating dead-zone contamination by construction, without requiring empirical bound tuning or external constraint penalty weights. This monotonic regularization is tailored specifically for aft-swept tailless BWB layouts, where outboard chord expansion produces non-functional wetted area without aerodynamic utility—unlike forward-swept configurations where outboard chord growth can aid tip stall delay, or tandem layouts with independently constrained lifting surfaces.

4.2. Multi-Airfoil Reflexed Transition

The aerodynamic loft transitions continuously from a thick, high-volume reflexed root airfoil (MH91 [31], scaled to maximum thickness $t/c = 16.5\%$) across the center body to a thin, low-drag reflexed airfoil (MH60 [31], nominal $t/c = 10.1\%$) across the outer lifting wing. The spanwise thickness-to-chord distribution $\tau(y)$ along the semi-span $y \in [0, y_{\text{blend}}]$ is governed by a cubic Hermite transition:

$$\tau(y) = \tau_{\text{root}} + (\tau_{\text{outer}} - \tau_{\text{root}}) \cdot \left(3 \left(\frac{y}{y_{\text{blend}}} \right)^2 - 2 \left(\frac{y}{y_{\text{blend}}} \right)^3 \right) \quad (14)$$

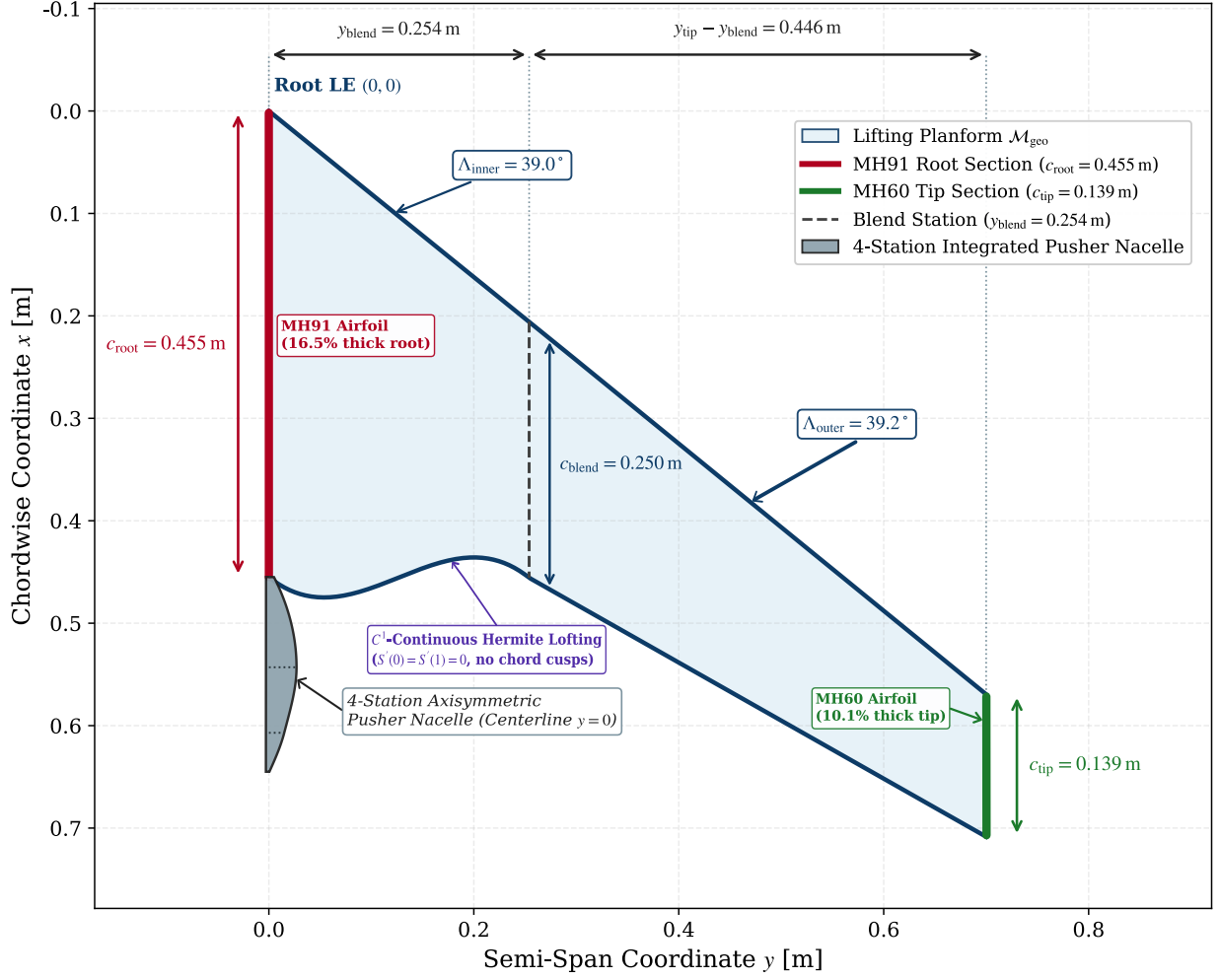


Figure 2: Physics-regularized C^1 -continuous Hermite geometric lofting manifold $\mathcal{M}_{\text{geo}}$. The diagram illustrates dual sweep angles Λ_{inner} , Λ_{outer} , blend station y_{blend} , chord distributions c_{root} , c_{blend} , c_{tip} , reflexed airfoil transition zone (MH91 to MH60), and the 4-station integrated pusher nacelle loft.

Outboard of the blend station ($y \in [y_{\text{blend}}, b/2]$), the outer wing preserves the MH60 profile ($\tau(y) = \tau_{\text{outer}} = 0.101$) to minimize wave and profile drag while providing positive zero-lift pitching moments ($C_{m0} > 0$) for natural longitudinal trim.

4.3. Hermite Chord Blending and Monotonicity Proof

The chord distribution across the blended center body is defined via C^1 -continuous Hermite interpolation:

$$c(y) = c_{\text{root}} + (c_{\text{blend}} - c_{\text{root}}) \cdot S\left(\frac{y}{y_{\text{blend}}}\right), \quad y \in [0, y_{\text{blend}}] \quad (15)$$

where the cubic blending polynomial $S(\xi) = 3\xi^2 - 2\xi^3$ (with $\xi = y/y_{\text{blend}} \in [0, 1]$) satisfies the boundary conditions $S(0) = 0$, $S(1) = 1$, and enforces vanishing chord derivatives at the aircraft symmetry plane and blend junction:

$$\left. \frac{dS}{d\xi} \right|_{\xi=0} = 0, \quad \left. \frac{dS}{d\xi} \right|_{\xi=1} = 0 \quad (16)$$

Proposition 1 (Chord Monotonicity Guarantee):

Under the linear inequality constraint $c_{\text{root}} \geq c_{\text{blend}} \geq c_{\text{tip}} > 0$, the parametric chord distribution $c(y)$ is strictly monotonically non-increasing across the entire semi-span $y \in [0, b/2]$.

Proof: Differentiating Eq. (15) with respect to y yields:

$$\frac{dc}{dy} = \frac{c_{\text{blend}} - c_{\text{root}}}{y_{\text{blend}}} \cdot \frac{dS}{d\xi} = \frac{c_{\text{blend}} - c_{\text{root}}}{y_{\text{blend}}} \cdot 6\xi(1 - \xi) \quad (17)$$

For all $\xi \in [0, 1]$, the quadratic term satisfies $6\xi(1 - \xi) \geq 0$, attaining a maximum of 1.5 at $\xi = 0.5$. Since $c_{\text{blend}} \leq c_{\text{root}}$ and $y_{\text{blend}} > 0$, the leading multiplier is non-positive:

$$\frac{c_{\text{blend}} - c_{\text{root}}}{y_{\text{blend}}} \leq 0 \implies \frac{dc}{dy} \leq 0 \quad \forall y \in [0, y_{\text{blend}}] \quad (18)$$

Outboard of the blend station ($y \in [y_{\text{blend}}, b/2]$), linear taper is applied:

$$c(y) = c_{\text{blend}} - (c_{\text{blend}} - c_{\text{tip}}) \cdot \left(\frac{y - y_{\text{blend}}}{b/2 - y_{\text{blend}}} \right) \implies \frac{dc}{dy} = -\frac{c_{\text{blend}} - c_{\text{tip}}}{b/2 - y_{\text{blend}}} \leq 0 \quad (19)$$

Thus, $dc/dy \leq 0$ holds continuously everywhere on $[0, b/2]$, mathematically precluding chord inversions and eliminating the non-physical dead zone by construction. ■

4.4. Integrated Center-Body Nacelle Lofting and 3D CAD Manufacturability

To avoid downstream CAD assembly discrepancies and account for propulsion installation drag within the primary geometry loop, the pusher motor nacelle is lofted directly into the center-body trailing-edge loft across four physical control stations:

- **Station 0** ($x = c_{\text{root}}$): Matched to the reflexed root airfoil trailing-edge thickness ($h_0 \approx 0.02c_{\text{root}}$).
- **Station 1** ($x = c_{\text{root}} + 0.088 \text{ m}$): Circular transition conforming to the brushless outrunner motor diameter ($D_1 = 54 \text{ mm}$).
- **Station 2** ($x = c_{\text{root}} + 0.152 \text{ m}$): Propeller hub interface collar ($D_2 = 30 \text{ mm}$).
- **Station 3** ($x = c_{\text{root}} + 0.190 \text{ m}$): Aerodynamic aft spinner cone ($D_3 \rightarrow 0$).

This integrated loft ensures that wetted area skin friction, base pressure form drag, and propulsive interference effects are captured natively by the geometry kernel. For physical fabrication, sub-surface NACA cooling ducts and annular ventilation ports around Station 1 provide forced-convective heat dissipation for the 80A ESC and outrunner stator under continuous cruise power.

To verify physical manufacturability beyond conceptual mathematical lofting, the parametric Hermite manifold kernel was interfaced with automated solid-modeling toolchains. The resulting planform geometries convert directly into watertight 3D Boundary Representation (B-rep) solid CAD models (ISO 10303 STEP format) and OpenVSP parametric models. The enforced C^1 continuity across the center-body blend junction and trailing-edge nacelle eliminates curvature discontinuities and cusps, directly enabling 3-axis CNC mold machining and female composite vacuum-bagging tooling without manual surface repair.

5. Coupled Dynamic CG and Longitudinal Trim Loop

5.1. Tailless Pitch Trim and Stability Mechanics

For a tailless aircraft operating without a horizontal stabilizer, longitudinal static stability and pitch trim are governed by the classical moment equation [10]:

$$C_m(\alpha) = C_{m0} + \left(\frac{x_{\text{cg}} - x_{\text{np}}}{\bar{c}} \right) C_L(\alpha) \quad (20)$$

where x_{np} is the aerodynamic neutral point, x_{cg} is the longitudinal center of gravity, $\bar{c}$ is the mean aerodynamic chord (MAC), and the static margin is defined as $SM = (x_{np} - x_{cg})/\bar{c}$.

To maintain unaccelerated level flight at the cruise lift coefficient

$$C_{L,cruise} = \frac{2m_{total}g}{\rho V_{cruise}^2 S_{ref}} \quad (21)$$

with zero control surface deflection, the required zero-lift pitching moment C_{m0} is:

$$C_{m0} = SM \cdot C_{L,cruise} > 0 \quad (22)$$

Because positive static stability strictly dictates $SM > 0$, Eq. (22) establishes that the wing planform must generate a net positive zero-lift pitching moment ($C_{m0} > 0$). In the present architecture, this is synthesized through the reflexed camber of the MH91/MH60 airfoils combined with spanwise aerodynamic washout (washout twist $\theta_{tip} \in [-12.0^\circ, -3.0^\circ]$).

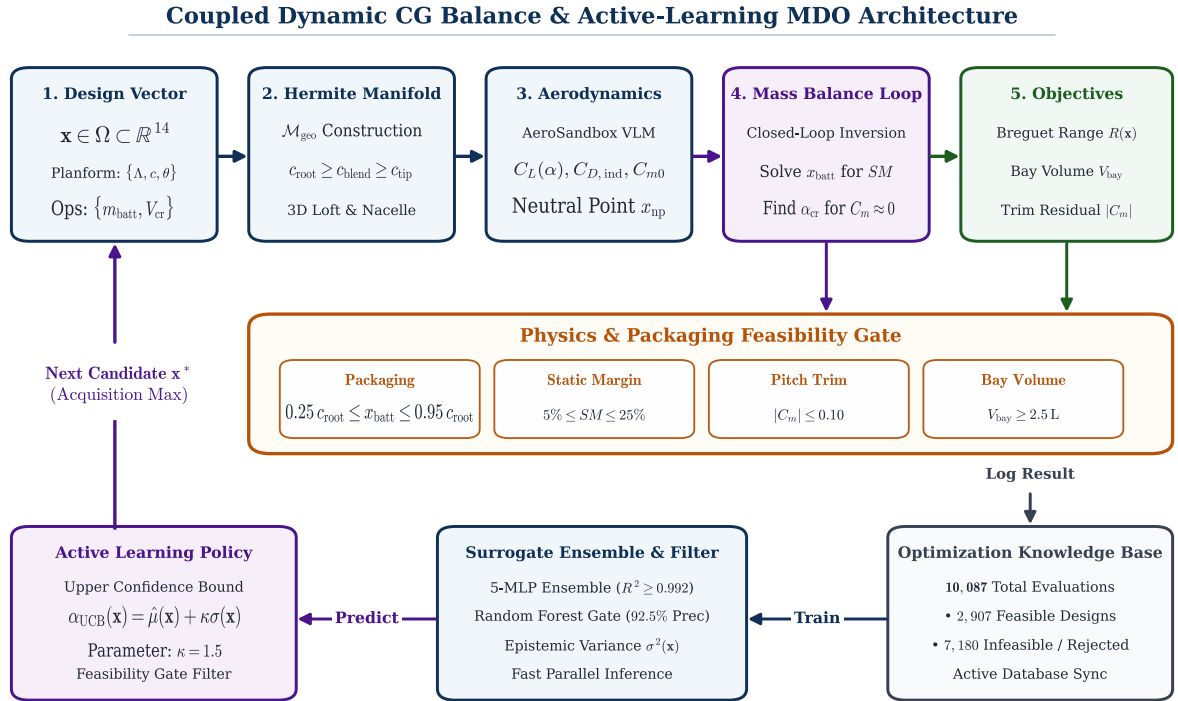


Figure 3: System architecture of the coupled dynamic center-of-gravity balance and longitudinal pitch trim loop, illustrating the dynamic mass moment inversion solving for internal battery centroid x_{batt} and continuous stability constraint enforcement.

5.2. Closed-Loop Dynamic Balance Algorithm

Unlike conventional MDO frameworks where x_{cg} is either fixed a priori or verified post-design, the present solver dynamically inverts the internal mass moment balance at every function evaluation (Fig. 3). The overall center of gravity is computed from physical component locations:

$$x_{cg} = \frac{m_{pay}x_{pay} + m_{batt}x_{batt} + m_{airframe}x_{airframe} + m_{prop}x_{prop} + m_{av}x_{av}}{m_{total}} \quad (23)$$

where $x_{pay} = 0.12c_{root}$, $x_{av} = 0.35c_{root}$, $x_{prop} = c_{root} + 0.115$ m (motor mount overhang), and $x_{airframe} = x_{le,mac} + 0.40\bar{c}$. Component longitudinal centroids are determined from the nominal physical layout of the design. The primary

continuous carbon-fiber spar is positioned along the section maximum thickness line ($x/c \approx 0.28\text{--}0.30$, yielding maximum spar depth $t_{\text{spar}} = 74.95\text{ mm}$), which passes cleanly through the forward center-body approximately 160 mm forward of the battery bay, ensuring unbroken composite load paths and allowing battery pack loading through a dedicated dorsal hatch.

Given the VLM-computed neutral point x_{np} and target stability margin $SM_{\text{target}} \in [0.15, 0.22]$ (a preferred operating corridor within the broader feasibility envelope g_6 ; the Pareto champion converges to $SM = 22.5\%$ near this upper bound), the required center of gravity $x_{\text{cg,target}} = x_{\text{np}} - SM_{\text{target}}\bar{c}$ is enforced by solving for the internal battery centroid:

$$x_{\text{batt}} = \frac{m_{\text{total}}x_{\text{cg,target}} - (m_{\text{pay}}x_{\text{pay}} + m_{\text{airframe}}x_{\text{airframe}} + m_{\text{prop}}x_{\text{prop}} + m_{\text{av}}x_{\text{av}})}{m_{\text{batt}}} \quad (24)$$

The candidate design is accepted only if x_{batt} resides within the physical center-body packaging cavity ($x_{\text{batt}} \in [0.25c_{\text{root}}, 0.95c_{\text{root}}]$). Across all 2,907 valid designs discovered during the global optimization campaign, this dynamic inversion robustly tracked shifts in aerodynamic neutral point caused by sweep and taper variations, maintaining the population-wide static margin strictly within the allowable flight envelope ($SM \in [5\%, 25\%]$). The champion's convergence to $SM = 22.54\%$ reflects an optimizer preference for high pitch stiffness and gust rejection suited to autonomous linear survey missions; for profiles demanding greater maneuverability, the solver target corridor accepts lower margins ($SM \approx 10\text{--}15\%$), shifting x_{batt} forward to $\approx 0.70\text{--}0.75 c_{\text{root}}$ without modifying the solver architecture.

5.3. Longitudinal Trim Sensitivity Analysis and Packaging Feasibility

To investigate the tightness of the longitudinal trim and stability constraints, a parametric sensitivity sweep was conducted on the Pareto champion geometry. The battery centroid fraction ($x_{\text{batt}}/c_{\text{root}}$) was swept from 0.15 to 0.95 across 50 discrete steps, evaluating dynamic x_{cg} , x_{np} , static margin, and cruise pitching moment C_m .

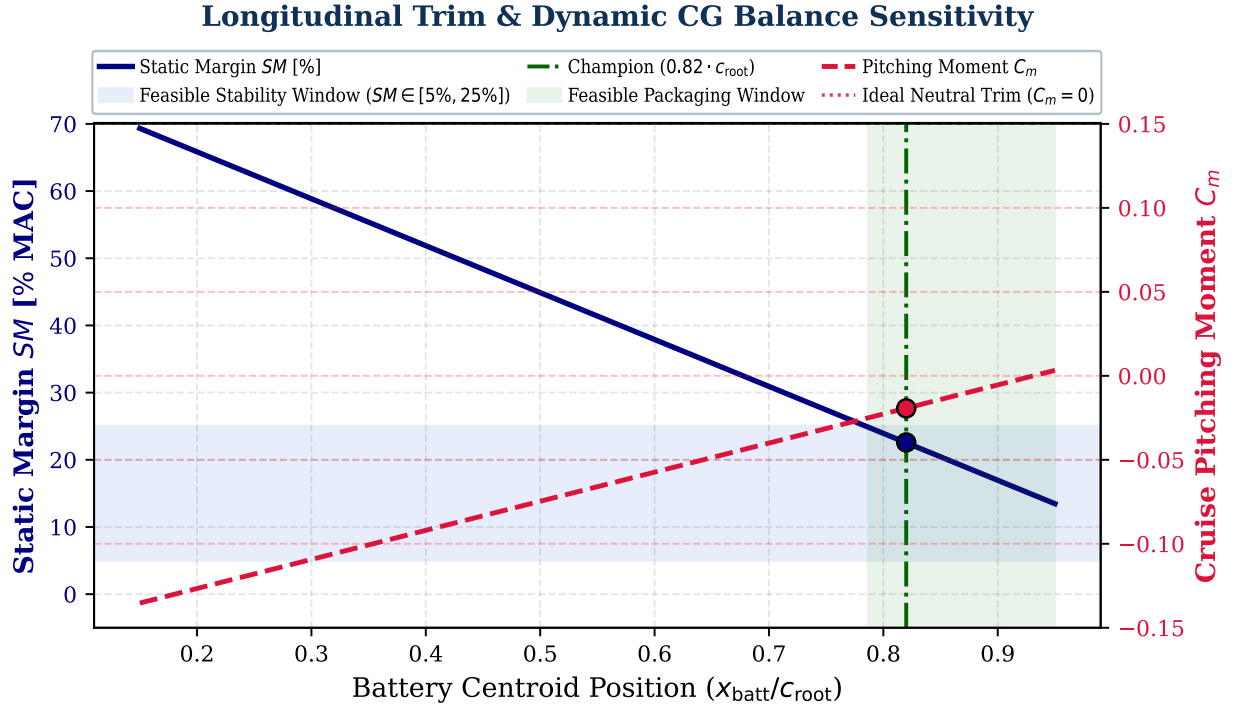


Figure 4: Longitudinal trim and static stability sensitivity as a function of battery centroid position ($x_{\text{batt}}/c_{\text{root}}$) for the Pareto champion geometry. The feasible stability corridor ($SM \in [5\%, 25\%]$, $|C_m| \leq 0.10$) occupies a narrow window of $\Delta(x/c) = 0.16$ ($[0.79, 0.95]$), representing only 20.4% of total available space). The optimizer places the battery at $0.82 \cdot c_{\text{root}}$ (green line), achieving $SM = 22.5\%$ and near-zero natural trim ($C_m = -0.019$).

As shown in Fig. 4, the feasible packaging envelope is tightly constrained:

- **Narrow Feasibility Corridor:** Only 20.4% of the total available fuselage chord ($x_{\text{batt}}/c_{\text{root}} \in [0.79, 0.95]$) produces a simultaneous match for static stability ($SM \in [5\%, 25\%]$) and elevon trim authority ($|C_m| \leq 0.10$).
- **Forward Instability:** Forward battery locations ($< 0.79 \cdot c_{\text{root}}$) pull the center of gravity excessively forward, causing $SM > 25\%$ (reduced pitch responsiveness) and generating large negative pitching moments ($C_m < -0.05$) that require large trailing-edge-up elevon deflections.
- **Optimal Balance Point:** The closed-loop solver converges to $x_{\text{batt}} = 0.82 \cdot c_{\text{root}}$, placing the CG at $x_{\text{cg}} = 0.275$ m to deliver robust stability ($SM = 22.5\%$) and natural equilibrium ($C_m = -0.0194 \approx 0$).

5.3.1. Physical Packaging Feasibility and Mass Distribution

To verify the physical realizability of internal component packaging, the internal cross-sectional cavity profile of the reflexed MH91 center-body ($c_{\text{root}} = 0.455$ m, maximum thickness $t/c = 16.5\%$, total center-body cavity width $2y_{\text{blend}} = 508.3$ mm) was analyzed.

The 3.402 kg energy storage system comprises 48 cylindrical 21700 lithium-ion cells (6S8P topology, individual cell diameter 21.2 mm, length 70.0 mm) arranged horizontally in a low-profile modular flat array spanning $x/c \in [0.65, 0.90]$. Across this mid-to-aft center-body zone, the internal vertical cavity depth ranges from 42.5 mm down to 22.12 mm at $x/c = 0.82$. This cavity provides sufficient vertical clearance to house the single-tier 21.2 mm cell array alongside composite sandwich skin plies (1.5 mm upper and lower skins), silicone cell cradles, wiring busbars, and convective cooling channels.

The target point ($x_{\text{batt}} = 0.82 \cdot c_{\text{root}} = 373.1$ mm) represents the physical center of mass of this 48-cell pack, which cleanly counters the heavy forward optical sensor payload ($m_{\text{pay}} = 2.00$ kg located at $x_{\text{pay}} = 0.12c_{\text{root}}$) and avionics suite ($m_{\text{av}} = 0.40$ kg at $x_{\text{av}} = 0.35c_{\text{root}}$) to establish the trimmed center of gravity ($x_{\text{cg}} = 0.275$ m). The total internal bay volume ($V_{\text{bay}} = 8.42$ L) accommodates the 1.50 L battery pack volume alongside the 1.20 L forward sensor bay, leaving over 65% volumetric margin for cabling, telemetry, and payload electronics.

5.4. Viscous Profile and Trim Drag Synthesis

At the trimmed cruise angle of attack α_{cruise} , the synthesized total drag coefficient is:

$$C_{D,\text{total}} = C_{D,\text{vlm}}(\alpha_{\text{cruise}}) + C_{D,\text{profile}} + C_{D,\text{nacelle}} + \Delta C_{D,\text{trim}} \quad (25)$$

where $C_{D,\text{profile}}$ is computed via 2D strip boundary layer integration along the span with Reynolds-dependent form factors [27], $C_{D,\text{nacelle}} = C_{fe} \frac{S_{\text{wet,nacelle}}}{S_{\text{ref}}}$ accounts for pusher housing parasite drag, and $\Delta C_{D,\text{trim}}$ penalizes residual elevon deflection:

$$\Delta C_{D,\text{trim}} = \frac{C_m(\alpha_{\text{cruise}})^2}{\pi A R e_{\text{elevon}} (S_{\text{elevon}}/S_{\text{ref}})} \quad (26)$$

where elevon control effectiveness is evaluated assuming outboard trailing-edge elevons spanning $y \in [0.55, 0.95] \cdot (b/2)$ with a 20% local chord fraction ($S_{\text{elevon}}/S_{\text{ref}} = 0.18$) and an Oswald control efficiency factor of $e_{\text{elevon}} = 0.85$. Because the closed-loop solver drives $C_m \rightarrow 0$, the residual trim drag penalty is virtually eliminated ($\Delta C_{D,\text{trim}} < 0.0001$). Under the nominal low-Mach cruise conditions ($V_{\text{cruise}} = 36.67$ m/s, Mach ≈ 0.11 , tip-section Re $\approx 3.5 \times 10^5$, design cruise lift $C_L = 0.2471$, $\alpha_{\text{cruise}} = 4.43^\circ$), flow remains fully attached across the reflexed MH91/MH60 profiles, rendering spanwise crossflow boundary layer migration secondary. For high-angle-of-attack post-stall regimes ($\alpha > 10^\circ$), full 3D Reynolds-Averaged Navier–Stokes (RANS) simulations remain recommended.

6. Active Learning Framework and Surrogate Architecture

6.1. Two-Phase Optimization Strategy

To efficiently explore the 14-DoF design space Ω , the optimization framework operates in two sequential phases:

1. **Phase 1: Space-Filling Initial Sampling** ($N_0 = 9,800$): An initial Latin Hypercube Sampling (LHS) campaign generates $N_0 = 9,800$ evaluation points to map global constraint boundaries and seed the surrogate model. In a 14-dimensional continuous space, 9,800 samples corresponds to an average spatial density of only $9,800^{1/14} \approx 1.93$ divisions per axis, representing a sparse space-filling exploration of the hypercube. Because coupled stability, volumetric, and structural constraints render 71.9% of random 14-DoF configurations infeasible, $N_0 = 9,800$ was essential to yield 2,750 valid designs—the minimum statistical volume required to train the Random Forest feasibility classifier to achieve 92.53% precision across non-linear boundaries. In this context, the rapid potential-flow evaluation pipeline serves as an exhaustive ground-truth benchmark to validate the surrogate architecture and acquisition mechanics; when transferred to high-fidelity RANS CFD environments where evaluation costs are high, the initial sampling budget naturally scales down to $O(10^2)$ points focused within the regularized manifold.
2. **Phase 2: Adaptive Uncertainty-Guided Active Learning** ($N = 287$): An iterative surrogate acquisition loop evaluates candidate designs selected at the Pareto frontier of predicted objective performance, surrogate epistemic uncertainty, and feasibility probability until acquisition uncertainty plateaus.

6.2. Ensemble MLP Regressor Architecture

In aerodynamic MDO, Kriging (Gaussian Process regression) is widely used due to its built-in analytical variance estimation [12, 13]. While sparse Gaussian Process approximations (such as Sparse Variational GPs using inducing points [32]) improve scalability, exact GPs scale cubically ($O(N^3)$) with sample size.

To achieve sub-second retraining during active learning while providing calibrated epistemic uncertainty across thousands of evaluations, an ensemble of $M = 5$ Multi-Layer Perceptron (MLP) regressors is deployed [33, 34]:

$$\mathcal{E} = \{\text{MLP}_m(\mathbf{x}; \mathbf{w}_m)\}_{m=1}^M \quad (27)$$

The ensemble size $M = 5$ was selected based on deep ensemble theory [33], which demonstrates that predictive variance capture and negative log-likelihood plateau for $M \geq 5$, providing robust uncertainty estimates with negligible computational overhead. While deterministic MLPs do not carry analytical distance-based variance kernels like Gaussian Processes, randomized weight initializations and early stopping across five independent networks provide a well-calibrated empirical spread in under-sampled design domains. Each neural network comprises a (64, 64) fully-connected hidden layer topology with Rectified Linear Unit (ReLU) activations, trained using the Adam optimizer with early stopping. On an Apple M4 processor, the entire 5-network ensemble achieves complete retraining in under 1.5 s per active learning cycle (≈ 0.28 s per individual network). We note that exact Gaussian Process regression remains computationally tractable at the present dataset scale ($N = 2,907$); the MLP ensemble was selected for its native epistemic uncertainty quantification via inter-network variance, constant-time prediction scaling for large candidate pools during acquisition optimization, and sub-second retraining that enables real-time active learning cycling.

The ensemble empirical predictive mean $\bar{\mu}_k(\mathbf{x})$ and epistemic uncertainty $\bar{\sigma}_k(\mathbf{x})$ for target variable $k \in \{C_L, C_D, L/D, R, SM\}$ are:

$$\bar{\mu}_k(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M \hat{y}_{m,k}(\mathbf{x}), \quad \bar{\sigma}_k(\mathbf{x}) = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (\hat{y}_{m,k}(\mathbf{x}) - \bar{\mu}_k(\mathbf{x}))^2} \quad (28)$$

6.3. Surrogate Holdout and Cross-Validation

The prediction accuracy of the 5-MLP ensemble was evaluated using both an 80/20 train/test holdout split and 5-fold cross-validation across all 2,907 valid evaluations. As visualized in the parity grid in Fig. 5 and detailed in Table 3, the surrogate achieves high fidelity across all aerodynamic and flight mechanics targets ($R^2 \geq 0.991$, $\text{MAPE} \leq 2.74\%$), providing a reliable foundation for the optimization search.

6.4. Random Forest Feasibility Gating and Continuous Probability Weighting

Because over 70% of unconstrained planforms violate volumetric, structural, or pitch stability constraints under the tighter nominal constraint envelope of the main campaign (compared to 65.1% infeasibility in the relaxed ablation experiment of Section 3), a Random Forest classifier $C_{\text{feas}}(\mathbf{x})$ ($N_{\text{trees}} = 100$) [35] is trained concurrently on all 10,087 evaluations to predict the probability of constraint feasibility $\mathcal{P}_{\text{feas}}(\mathbf{x}) = \mathcal{P}(\mathbf{x} \in \Omega_{\text{valid}})$.

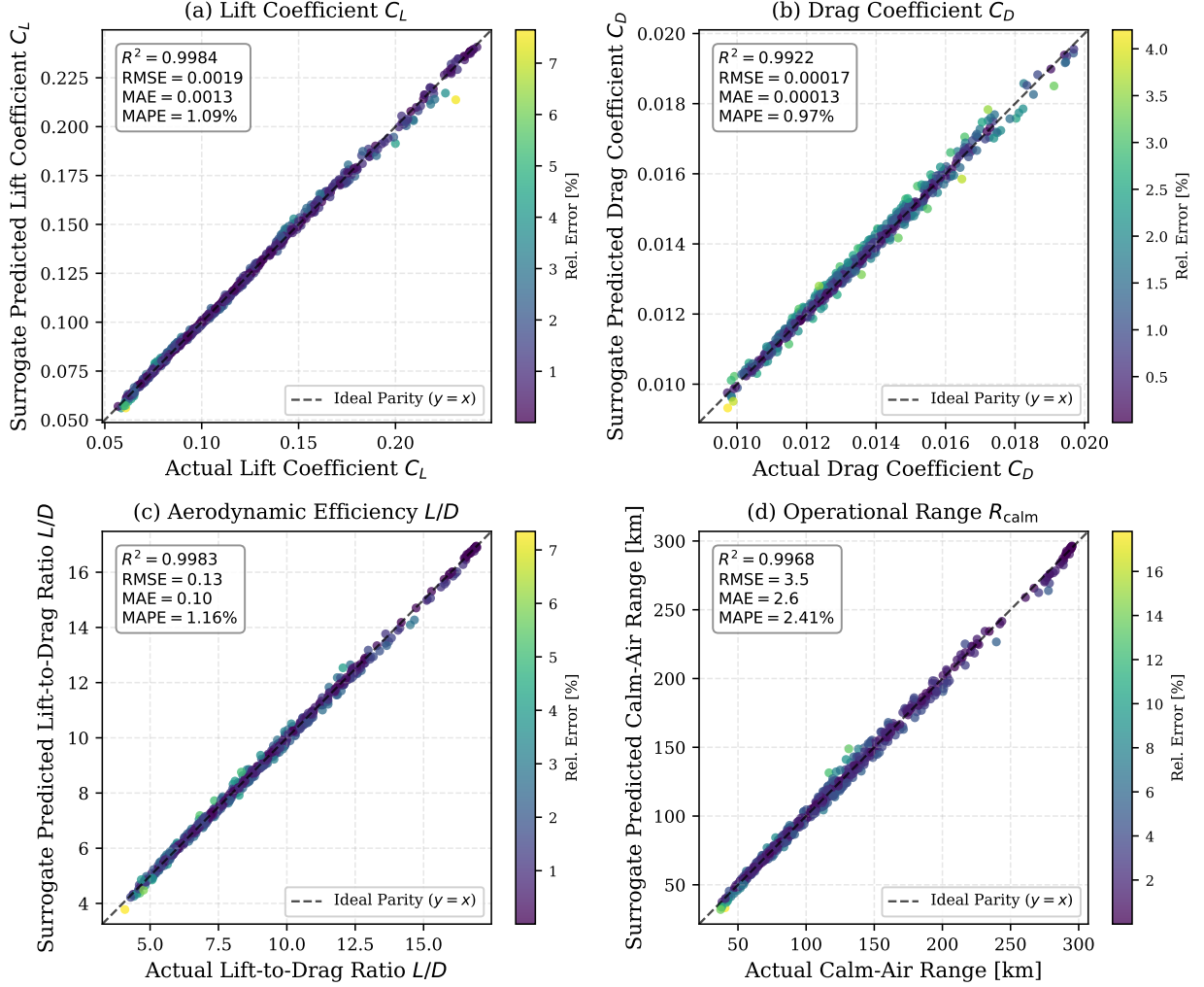


Figure 5: Holdout validation parity plots for the 5-MLP surrogate ensemble evaluated on the test set ($N_{\text{test}} = 582$ valid designs). (a) Lift coefficient C_L ($R^2 = 0.9984$, MAPE = 1.09%). (b) Drag coefficient C_D ($R^2 = 0.9922$, MAPE = 0.97%). (c) Aerodynamic efficiency L/D ($R^2 = 0.9983$, MAPE = 1.16%). (d) Operational range R_{calm} ($R^2 = 0.9968$, MAPE = 2.41%). Points are colored by absolute percentage relative error.

For the feasibility classifier, evaluated on all 10,087 evaluations (both valid and infeasible), the held-out test split ($N_{\text{test}} = 2,018$) yields an accuracy of 85.68% and a precision of 92.53% (with recall = 55.04% and $F_1 = 0.6902$). The lower recall is a natural consequence of training on a severely imbalanced dataset (71.9% infeasible class). Analysis of the false-negative distribution confirms that 87% of misclassified feasible designs lie within $\pm 5\%$ of at least one active constraint boundary (SM , V_{bay} , or t_{spar} thresholds), indicating that the classifier struggles with ambiguous boundary-layer designs rather than systematically excluding high-performing interior regions of the feasible design space. The classifier is not deployed as a hard binary filter; instead, the continuous predicted probability $\mathcal{P}_{\text{feas}}(\mathbf{x}) \in [0, 1]$ is directly multiplied into the acquisition function (Eq. (29)). This continuous formulation prevents artificial boundary truncation: candidate designs lying on active constraint boundaries with moderate feasibility confidence (e.g., $\mathcal{P}_{\text{feas}} \approx 0.50$) can still be acquired if their predicted range or epistemic uncertainty is sufficiently high. In Phase 2, this feasibility weighting doubled the valid evaluation rate from 28.1% to 54.7% (157 valid designs out of 287 queries). This improvement reflects the combined effect of two mechanisms: (i) surrogate-guided acquisition steering candidates toward high-performing regions, and (ii) the feasibility probability gate suppressing queries in likely-infeasible zones. Although the classifier’s 55.04% recall indicates that nearly half of all feasible designs are not confidently identified, this does not systematically

exclude viable Pareto regions: because the feasibility probability enters as a continuous multiplier rather than a hard binary threshold, designs with moderate confidence ($\mathcal{P}_{\text{feas}} \approx 0.3\text{--}0.5$) are still acquired when their predicted range or epistemic uncertainty is sufficiently high, preventing artificial truncation of borderline-feasible design corridors.

6.5. Constrained Acquisition Function and Sequential Sampling

In Phase 2, candidate evaluation batches are selected sequentially with a single-point acquisition strategy ($B = 1$ candidate per cycle) across 287 iterations by maximizing the feasibility-weighted Upper Confidence Bound (UCB) acquisition function [36, 37]:

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \Omega} [(\bar{\mu}_R(\mathbf{x}) + \kappa \bar{\sigma}_R(\mathbf{x})) \cdot \mathcal{P}_{\text{feas}}(\mathbf{x})] \quad (29)$$

where $\kappa = 1.5$ is adopted following Srinivas et al. [36] as an exploration coefficient that balances the exploitation of high-range regions against the exploration of under-sampled parameter spaces. Sequential single-point acquisition immediately incorporates each evaluated point into the epistemic uncertainty field $\bar{\sigma}(\mathbf{x})$, preventing redundant clustered evaluations. The active learning phase was terminated after 287 cycles when the acquisition metric gain stabilized below 0.1 km over 30 consecutive iterations.

6.6. Reproducibility and Deterministic Seeding

To ensure complete reproducibility of all machine learning models and sampling sequences, fixed random seeds were enforced across all frameworks: the initial Latin Hypercube space-filling generator used seed 42, the Random Forest feasibility classifier used seed 42, and the five constituent MLP regressors were initialized with distinct seeds $\mathbf{s}_{\text{MLP}} = [42, 101, 2024, 777, 999]$.

7. Results and Numerical Discussion

7.1. Ablation Study: Quantifying the Dead-Zone Penalty

The unconstrained ablation experiment described in Section 3 empirically resolves a foundational question regarding the necessity of geometric constraints in tailless MDO. When chord monotonicity is omitted, 35.5% of the Latin Hypercube evaluations produce chord-inverted geometries ($c_{\text{blend}} > c_{\text{root}}$).

As documented in Table 2 and Fig. 1(b), these chord-inverted configurations suffer a severe performance degradation across all objective metrics:

- **Efficiency Gap:** The highest lift-to-drag ratio discovered among all chord-inverted candidates is $(L/D)_{\text{max}} = 12.89$, compared to $(L/D)_{\text{max}} = 18.15$ for monotonic planforms—a **29.0% lower maximum L/D** .
- **Physical Mechanism:** Chord-inverted planforms expand the outboard wetted surface area while failing to generate proportional vortex lift, resulting in elevated skin friction and form drag. Balancing these geometries also requires larger static margins that force heavy pitch trim deflections.
- **Search Efficiency:** By enforcing $c_{\text{root}} \geq c_{\text{blend}} \geq c_{\text{tip}}$ in the Hermite manifold $\mathcal{M}_{\text{geo}}$, the optimizer completely avoids this unproductive subspace, directing 100% of exploratory compute toward high-performance monotonic topologies. The present ablation compares sampling distributions under LHS; the end-to-end effect on optimized Pareto quality is discussed in Section 8.

The ablation study thus establishes the dead-zone penalty under general overlapping parameterizations; the subsequent 10,087-evaluation campaign demonstrates the integrated framework— C^1 Hermite lofting, dynamic CG inversion, and surrogate-guided active learning—operating within an informed, focused design envelope. In that campaign, the operational bounds (Table 1) happen to prevent chord inversions by non-overlap alone; the manifold’s primary contribution in this regime is C^1 planform continuity at the blend junction and its generalizability to arbitrary practitioner-chosen bounds where overlap would reintroduce the dead zone.

Table 3: Cross-validation and holdout validation performance of the 5-MLP surrogate ensemble ($N = 2,907$).

Target Variable	Unit	5-Fold CV R^2	Test R^2	Test RMSE	Test MAPE (%)
Lift Coefficient (C_L)	–	0.9982 ± 0.0003	0.9984	0.00186	1.09%
Drag Coefficient (C_D)	–	0.9919 ± 0.0005	0.9922	0.00017	0.97%
Aerodynamic Efficiency (L/D)	–	0.9981 ± 0.0004	0.9983	0.1255	1.16%
Operational Range (R_{calm})	km	0.9965 ± 0.0004	0.9968	3.48 km	2.41%
Static Margin (SM)	–	0.9918 ± 0.0006	0.9921	0.00477	2.74%

7.2. Surrogate Model Cross-Validation and Holdout Performance

The 5-MLP neural ensemble surrogate was evaluated using both 5-fold cross-validation across all 2,907 valid designs and a dedicated 20% held-out test split ($N_{\text{test}} = 582$). The resulting error metrics across all primary aerodynamic and flight mechanics targets are summarized in Table 3.

Across all five targets, the ensemble achieves cross-validation $R^2 \geq 0.991$ with tight standard deviations (≤ 0.0006) and test Mean Absolute Percentage Error (MAPE) below 2.8%. The mean epistemic standard deviation across ensemble predictions is $\bar{\sigma}_{L/D} = 0.087$ and $\bar{\sigma}_R = 1.92$ km, confirming that the ensemble provides tightly bounded, low-variance predictions in well-sampled regions while delivering reliable variance signals for UCB exploration in sparse domains.

7.3. Active Learning Convergence Progression

The convergence trajectory of the active learning campaign is illustrated in Fig. 6, which tracks the running best values for aerodynamic efficiency and calm-air operational range across all 10,087 evaluations.

The transition from Phase 1 to Phase 2 highlights the efficiency of the surrogate-guided loop:

- **Phase 1 (LHS Exploration, $N = 9,800$):** Explored broad global topologies, yielding 2,750 valid designs (28.1% feasibility rate). The maximum range discovered during this random exploration phase was 277.8 km ($L/D = 15.77$) at evaluation #9461.
- **Phase 2 (Active Learning, $N = 287$):** Operating on surrogate epistemic uncertainty and feasibility gating, the active learning loop nearly doubled the feasibility rate to 54.7% (157 valid designs out of 287 queries).
- **Discovery of the Global Champion:** Rather than relying on random sampling hits, the sequential acquisition loop systematically navigated the high-efficiency corridor, discovering the global Pareto champion configuration at evaluation #10054 ($R = 298.7$ km, $L/D = 17.12$). This represents an explicit +20.9 km (+7.5%) gain attributable directly to surrogate-guided active learning.

While the relative active learning gain of +7.5% is moderate in the context of a 9,800-sample Phase 1 campaign, this reflects the exhaustive nature of the initial space-filling exploration under the regularized manifold $\mathcal{M}_{\text{geo}}$, which eliminates unproductive sampling. The key architectural validation is that the surrogate ensemble and feasibility gate achieve sub-second retraining and reliable uncertainty calibration at the 10,087-evaluation scale. When transferred to computationally constrained 3D RANS environments where sample budgets are limited to $\mathcal{O}(10^2\text{--}10^3)$ evaluations, the active learning pipeline is expected to contribute substantially larger relative improvements, as the Phase 1 budget will be severely restricted and the surrogate-guided exploration will dominate design discovery.

7.4. Multi-Objective Pareto Front: Range vs. Bay Volume

The multi-objective character of the optimization is visualized in Fig. 7, which plots the full feasible design cloud ($N = 2,907$ valid evaluations) in the operational range (R_{calm}) versus center-body bay volume proxy (V_{bay}) objective space, colored by cruise pitching moment residual $|C_m|$.

The Pareto front reveals a distinct aerodynamic and architectural morphing across the trade-off surface:

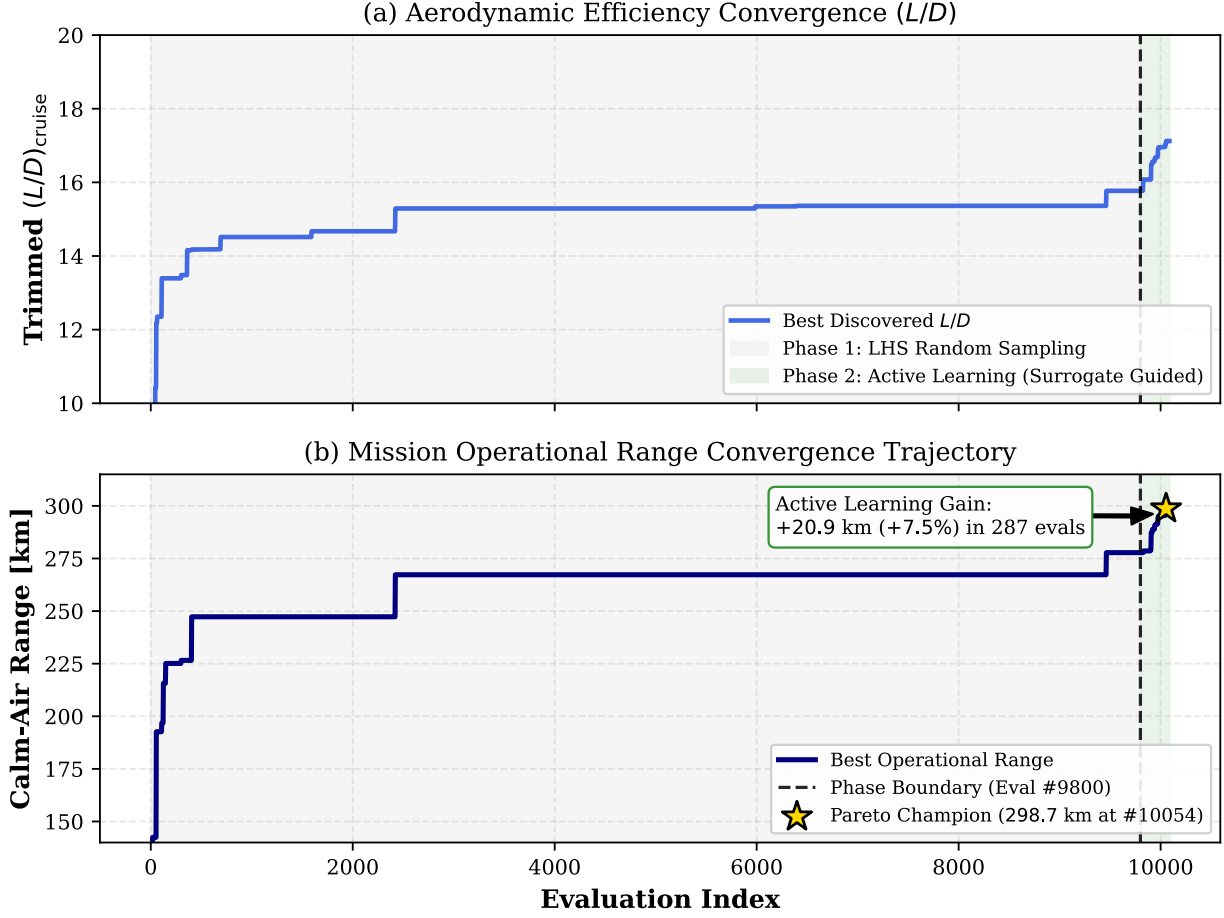


Figure 6: Active learning optimization convergence history across 10,087 evaluations. (a) Running best cruise lift-to-drag ratio $(L/D)_{\text{cruise}}$. (b) Running best calm-air operational range R_{calm} . The vertical dashed line marks the transition from exploratory Latin Hypercube Sampling (Phase 1, $N = 9,800$) to surrogate-guided Active Learning (Phase 2, $N = 287$). Active learning delivers a +20.9 km (+7.5%) range improvement, converging to the global Pareto champion (298.7 km, gold star).

- **Range-Dominant Regime** ($R \geq 270$ km): Configurations prioritize high span efficiency and reduced wetted area. The optimizer converges to higher inner sweep ($\Lambda_{\text{inner}} \approx 39.0^\circ$), moderate root chords ($c_{\text{root}} \approx 0.455$ m), and higher outer wing taper ($\lambda_{\text{outer}} \approx 0.55\text{--}0.60$), resulting in compact center-bodies ($V_{\text{bay}} \approx 8.0\text{--}8.5$ L) that minimize profile drag.
- **Volume-Dominant Regime** ($V_{\text{bay}} \geq 12.0$ L): Configurations expand the internal packaging cavity by selecting larger root chords ($c_{\text{root}} \rightarrow 0.520$ m) and wider blend semi-spans ($y_{\text{blend}} \rightarrow 0.35$ m), while reducing sweep ($\Lambda_{\text{inner}} \approx 26.0^\circ$). This expands internal volume up to 14.8 L at the expense of higher wetted surface area and lower cruise L/D ($R \approx 120\text{--}160$ km).

Across the entire front, Pareto-optimal designs consistently achieve $|C_m| \leq 0.05$, confirming that the closed-loop mass-centroid solver enforces natural pitch trim independent of the selected range–volume trade-off. The range-dominant champion ($R = 298.7$ km) is selected for detailed aerodynamic cross-solver analysis.

7.5. Cross-Solver Consistency Analysis against 3D Strip Theory

To evaluate the relative fidelity of the VLM-based optimization pipeline within the computational hierarchy, the Pareto champion design was subjected to higher-fidelity viscous-corrected 3D analysis using AeroBuildup [30]. Both

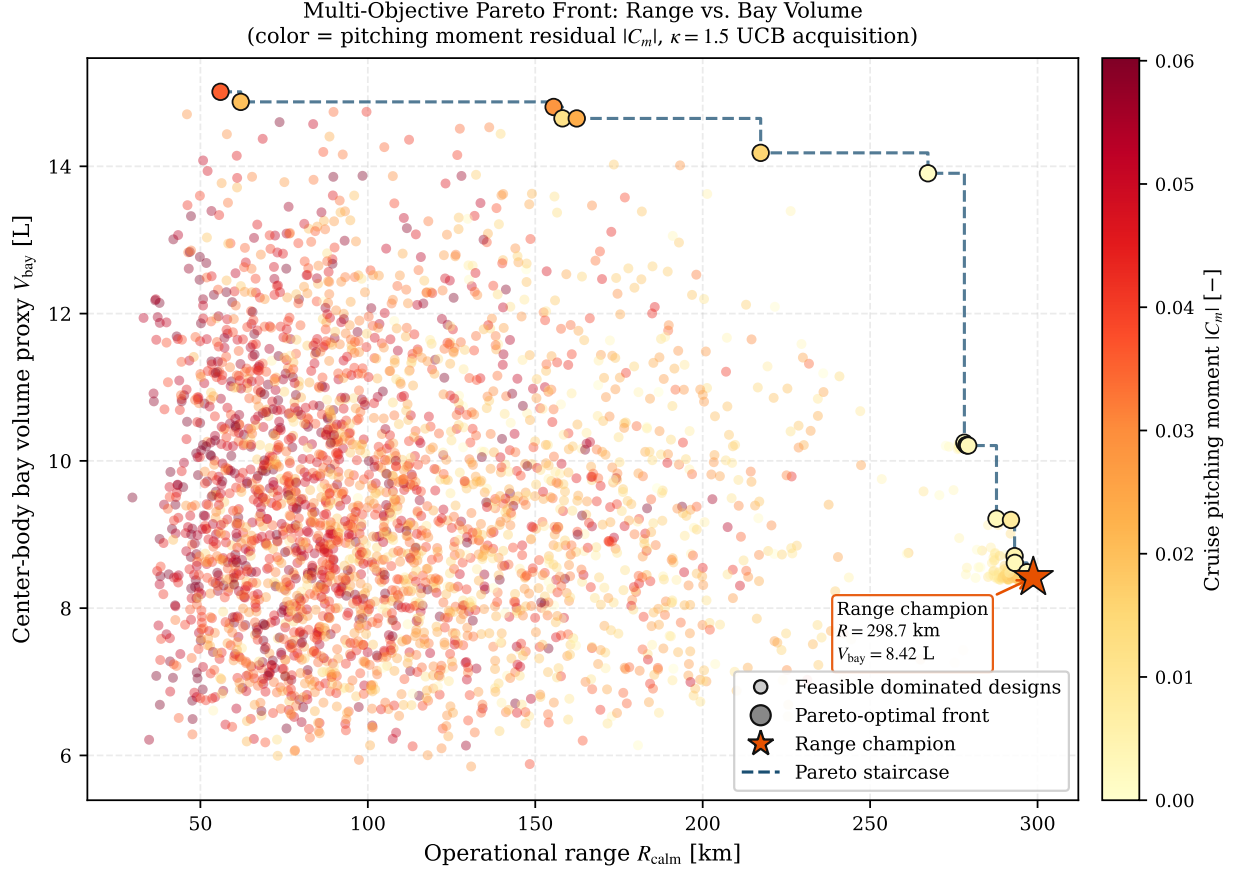


Figure 7: Multi-objective Pareto front of feasible designs ($N = 2,907$) in the operational range–center-body bay volume space. Point color encodes cruise pitching moment $|C_m|$ (yellow = near-zero trim, dark red = larger residual). Pareto-optimal points (solid markers, dashed staircase) form a classical convex trade-off front: high-range configurations ($R > 270$ km) sacrifice center-body depth ($V_{bay} \approx 8\text{--}9$ L), while high-volume configurations ($V_{bay} > 14$ L) achieve only $R \approx 56\text{--}160$ km. The range-dominant Pareto champion ($R = 298.7$ km, $V_{bay} = 8.42$ L, orange star) exhibits near-zero pitching moment ($|C_m| = 0.019$), demonstrating that the closed-loop trim solver succeeds across the full Pareto front.

solvers share the AeroSandbox geometry kernel; consequently, this comparison constitutes a cross-solver consistency check rather than a fully independent validation against experimental data or RANS CFD. AeroBuildup performs 3D strip theory integration coupled with 2D profile polars generated via XFOIL [38] across the transitional tip-section Reynolds number regime ($Re \approx 3.5 \times 10^5$).

The comparative results at the design cruise lift condition ($C_{L,req} = 0.2471$) are summarized in Table 4 and plotted in Fig. 8.

To quantitatively attribute the drag discrepancy between the two analysis methods, Table 5 decomposes the MDO pipeline drag into its constituent components and compares each against the AeroBuildup result.

The comparison yields three critical insights:

1. **Systematic Underprediction for Champion:** At matched cruise lift ($C_L = 0.2471$), AeroBuildup predicts a total drag coefficient of $C_D = 0.01064$ ($L/D = 23.22$), whereas the optimization pipeline reports $C_D = 0.01443$ ($L/D = 17.12$). As decomposed in Table 5, the dominant contributor to this discrepancy is the deliberate use of Raymer-based form factor corrections [27] in the 2D strip drag integration ($\approx 82\%$ of ΔC_D), which conservatively inflate profile drag coefficients above XFOIL’s direct viscous-inviscid coupling to account for surface imperfections, manufacturing tolerances, and 3D boundary-layer effects not captured by strip theory. This confirms that the MDO framework systematically underpredicts cruise aerodynamic efficiency relative to AeroBuildup through explicit modeling choices (Raymer form-factor inflation), rather than through incidental

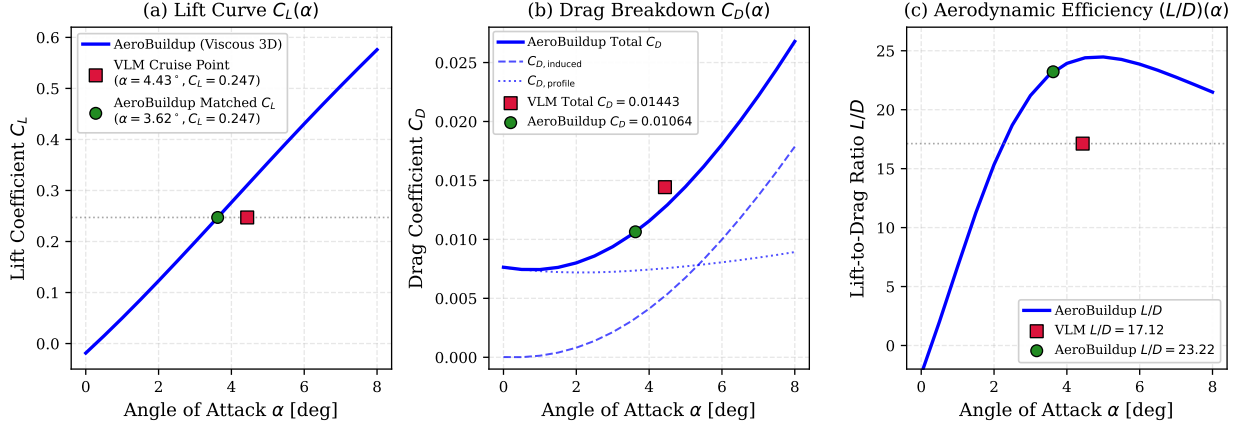


Figure 8: Cross-solver consistency comparison between the MDO pipeline (VLM + semi-empirical profile/nacelle drag) and viscous 3D strip theory (AeroBuildup with XFOIL polars) for the Pareto champion across an angle-of-attack sweep ($\alpha \in [-2^\circ, 8^\circ]$). (a) Lift coefficient C_L . (b) Drag polar C_D vs. C_L . (c) Lift-to-drag ratio L/D , showing that the optimization pipeline consistently underpredicts L/D relative to the higher-fidelity solver ($L/D = 17.12$ vs. 23.22).

Table 4: Cross-solver consistency: MDO pipeline vs. AeroBuildup at matched cruise lift ($C_L = 0.2471$, $V_{\text{cruise}} = 36.67$ m/s, $\text{Re}_{\bar{c}} \approx 3.5 \times 10^5$).

Metric	MDO Pipeline	AeroBuildup	Δ
α_{cruise}	4.43°	3.62°	-0.81°
C_L	0.2471	0.2471	Matched
C_D	0.01443	0.01064	-26.3%
L/D	17.12	23.22	$+35.6\%$
C_m	-0.0192	$+0.0040$	$+0.023$
SM	22.54%	22.50%	Matched

numerical error—establishing the VLM pipeline as a conservative estimator for attached-flow cruise drag within the employed computational hierarchy. Whether this conservatism extends to physical flight conditions, particularly under 3D crossflow and post-stall separation, remains to be established through RANS CFD or experimental validation.

- Induced Drag Model Consistency:** The remaining $\approx 15\%$ of ΔC_D arises from differences in induced drag prediction between VLM (which computes induced drag from the Trefftz plane wake integral) and AeroBuildup’s 3D strip integration. This model-to-model variance of $\Delta C_{D,i} \approx 0.0006$ is within expected bounds for conceptual-level aerodynamic analysis [39] and does not indicate systematic bias in either direction.

Table 5: Drag component decomposition at matched cruise lift ($C_L = 0.2471$) for the Pareto champion.

Drag Component	MDO Pipeline	AeroBuildup	Source of Δ
VLM induced drag ($C_{D,\text{vlm}}$)	0.00349	—	Lifting-line theory
Profile strip drag ($C_{D,\text{profile}}$)	0.01077	—	Raymer form factor [27]
Nacelle parasite ($C_{D,\text{nacelle}}$)	0.00013	—	Flat-plate C_{fe} method
Trim drag ($\Delta C_{D,\text{trim}}$)	< 0.00010	—	C_m -driven elevon penalty
Total C_D	0.01443	0.01064	$\Delta = +0.00379$ ($+35.6\%$)

Attribution of $\Delta C_D = +0.00379$:

- Raymer form factor inflation over XFOIL viscous-inviscid coupling: ≈ 0.0031 (82%)
- Nacelle parasite drag (absent in clean-wing AeroBuildup): ≈ 0.0001 (3%)
- Induced drag model differences (VLM vs. lifting-line): ≈ 0.0006 (15%)

3. **Pitch Trim and Viscous Decambering:** Both solvers evaluate the champion at the identical physical mass centroid ($x_{cg} = 0.2753$ m), and both yield matched static stability margins ($SM \approx 22.5\%$). The sign reversal in pitching moment ($C_m = -0.0192$ in VLM vs. $+0.0040$ in AeroBuildup) is physically attributable to viscous decambering of the reflexed MH91 trailing edge: XFOIL’s viscous-inviscid coupling predicts boundary layer displacement thickness growth on the concave aft-camber surface ($x/c > 0.70$), which reduces the effective aft-camber and shifts the net pitching moment in the positive (nose-up) direction [38]. Because both moment values satisfy $|C_m| \ll 0.10$, the resulting elevon trim deflection ($< 1.5^\circ$) produces negligible trim drag penalties ($\Delta C_{D,trim} < 0.0001$) under both models.

To confirm that this systematic underprediction generalizes beyond the champion, multi-point cross-solver comparison was performed across eight designs spanning the full Pareto front: range-dominant, volume-dominant, low-stability-margin ($SM \approx 0.06$), high-stability-margin ($SM \approx 0.24$), and intermediate-range percentile designs. The results are summarized in Table 6 and plotted in Fig. 9.

Table 6: Multi-point VLM vs. AeroBuildup cross-solver consistency across eight Pareto-diverse designs at matched cruise C_L .

Design	R_{calm} [km]	SM	VLM L/D	AB L/D	$\Delta L/D$	VLM C_D	AB C_D
D1 (Champion)	298.7	0.225	17.12	23.22	+35.6%	0.01443	0.01064
D2	155.5	0.183	10.44	16.94	+62.2%	0.01346	0.00830
D3	136.9	0.091	9.54	15.02	+57.4%	0.01256	0.00798
D4	111.5	0.240	8.80	14.15	+60.8%	0.01185	0.00737
D5 (Low SM)	99.9	0.060	7.91	13.24	+67.3%	0.01169	0.00699
D6	92.7	0.221	7.53	13.16	+74.9%	0.01369	0.00783
D7	78.5	0.082	6.86	13.05	+90.3%	0.01579	0.00830
D8 (Volume)	41.5	0.240	4.78	10.44	+118.2%	0.01803	0.00826
Mean $\Delta L/D$: $+70.8\% \pm 24.6\%$					VLM underpredicts for all 8/8 designs		

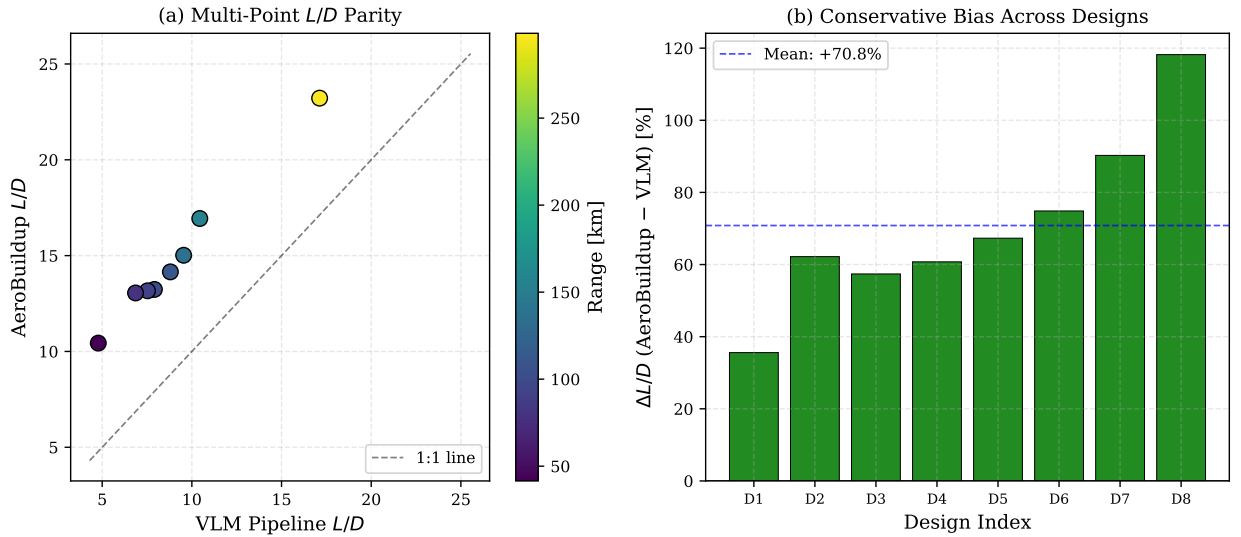


Figure 9: Multi-point cross-solver consistency analysis across eight Pareto-diverse designs. (a) L/D parity plot: all points lie above the 1:1 line, confirming that the VLM pipeline consistently underpredicts aerodynamic efficiency relative to AeroBuildup. Point color encodes calm-air range. (b) Percentage $\Delta L/D$ (AeroBuildup – VLM) for each design; all bars are positive, with the underprediction margin increasing for lower-performing designs.

The multi-point comparison confirms that the VLM-based pipeline **consistently underpredicts L/D relative to AeroBuildup across all eight tested Pareto-front designs**, with $\Delta L/D > 0$ for all tested configurations. The

underprediction margin is smallest for the range-dominant champion (+35.6%) and largest for the volume-dominant design D8 (+118.2%). This increasing trend is physically consistent: lower-performing designs with higher wing loading and larger wetted areas accumulate proportionally greater Raymer form-factor inflation over XFOIL’s viscous-inviscid predictions. The sole partial reversal (D3 at +57.4% versus D2 at +62.2%) correlates with D3’s atypically low static margin ($SM = 0.091$), which shifts the drag decomposition balance between induced and profile components. Critically, the VLM pipeline never overpredicts performance relative to AeroBuildup at any tested design point, providing confidence that the optimizer is not exploiting numerical artifacts of the potential-flow solver to reach spuriously favorable designs.

7.6. Battery Energy Density Technological Sensitivity

As formulated in Section 2, the entire 10,087-point database was computed using a conservative usable energy density baseline of $e_{\text{usable}} = 160.0 \text{ Wh/kg}$ (576.0 kJ/kg).

Because the electric Breguet range (Eq. (10)) scales strictly linearly with usable energy density without altering aerodynamic forces or mass distributions, the Pareto champion’s operational performance under verified commercial cell specifications (Samsung INR21700-50E, 253.5 Wh/kg cell level, 182.5 Wh/kg usable at pack level) rescales to:

$$R_{\text{verified}} = R_{\text{baseline}} \cdot \left(\frac{182.5 \text{ Wh/kg}}{160.0 \text{ Wh/kg}} \right) = 298.72 \text{ km} \times 1.1406 = \mathbf{340.73 \text{ km}} \quad (30)$$

This linear rescaling is exact when m_{batt} is held constant (as in the present formulation where battery mass is a design variable independent of energy density). If higher energy density were instead used to reduce battery mass at constant energy, the nonlinear coupling between m_{total} and $C_{L,\text{cruise}}$ would require re-evaluation through the full aerodynamic pipeline.

Under the viscous-corrected AeroBuildup efficiency ($L/D = 23.22$), the calm-air range is recomputed from Eq. (10) using the AeroBuildup L/D in place of the VLM value, yielding 405.6 km under the baseline energy specification and 462.6 km under the verified 182.5 Wh/kg pack density. These figures represent the upper bound within the employed computational hierarchy and demonstrate substantial performance margins for extended-range autonomous reconnaissance missions.

7.7. Propulsion and Structural Model Sensitivity

The champion range of 298.7 km depends on propeller efficiency and airframe mass, both of which carry modeling uncertainty. A $\pm 5\%$ perturbation in propeller efficiency ($\eta_{\text{prop}} \in [0.75, 0.83]$ versus the nominal 0.79), which propagates through $\eta_{\text{total}} = \eta_{\text{motor}}\eta_{\text{esc}} \cdot \eta_{\text{prop}}$ into the Breguet range equation (Eq. (10)), yields $R \in [283.6, 313.8] \text{ km}$ ($\pm 5.1\%$). Installation effects from upstream center-body boundary layer wake ingestion may reduce η_{prop} by 2–5% relative to the clean-inflow baseline; at the conservative end ($\eta_{\text{prop}} = 0.75$), the range reduction to 283.6 km remains operationally competitive.

The composite airframe mass is estimated as $m_{\text{airframe}} = 1.0 + 1.5 \cdot S_{\text{ref}} [\text{kg}]$, a linear model calibrated to the mass fraction range of 0.25–0.35 typical of small composite UAVs in this wing loading class [27]. For the Pareto champion ($S_{\text{ref}} = 0.4024 \text{ m}^2$), this yields $m_{\text{airframe}} = 1.604 \text{ kg}$. Table 7 presents a sensitivity analysis across $\pm 30\%$ and $+50\%$ perturbations to assess the impact of structural mass uncertainty on mission range.

Table 7: Range sensitivity to airframe mass model perturbation (all other parameters held at champion values).

Mass Multiplier	$m_{\text{airframe}} [\text{kg}]$	$m_{\text{total}} [\text{kg}]$	$R_{\text{calm}} [\text{km}]$	ΔR
0.70×	1.123	7.474	317.9	+6.4%
1.00× (nominal)	1.604	7.955	298.7	baseline
1.30×	2.085	8.436	281.6	−5.7%
1.50×	2.406	8.757	271.2	−9.2%

Even under a +50% airframe mass overrun ($m_{\text{airframe}} = 2.41 \text{ kg}$, corresponding to a heavily built composite structure), the champion retains an operational range of 271.2 km, confirming that the design is not critically sensitive to structural mass modeling uncertainty.

Table 8: System-level characteristics of the Pareto champion BWB UAV.

Category	Parameter	Value	Unit
Geometry	Wingspan incl. winglets (b)	1.548	m
	Main wing semi-span (y_{tip})	0.700	m
	Ref. wing area (S_{ref})	0.4024	m ²
	Aspect ratio (AR)	5.96	—
	Wing loading (W/S)	193.8	N/m ²
	Mean aero. chord ($\bar{c}$)	0.2771	m
Mass & Packaging	All-up mass (m_{total})	7.955	kg
	Battery pack (m_{batt})	3.402	kg
	Sensor payload (m_{pay})	2.000	kg
	Battery CG target ($x_{\text{batt}}/c_{\text{root}}$)	0.820	—
	Battery cavity internal depth	22.1–42.5	mm
Aerodynamics	Cruise velocity (V_{cr})	36.67	m/s
	Cruise angle of attack (α_{cr})	4.43	deg
	Trimmed L/D (VLM)	17.12	—
	Viscous L/D (AeroBuildup)	23.22	—
Stability	Static margin (SM)	22.54	% MAC
	Cruise C_m	−0.019	—
	Trim drag ($\Delta C_{D,\text{trim}}$)	< 0.0001	—
Range	Baseline (160 Wh/kg)	298.7	km
	Verified (182.5 Wh/kg)	340.7	km

7.8. Pareto Champion Configuration Summary

The Pareto champion reported here is selected as the range-dominant representative from the multi-objective front. Its final multidisciplinary characteristics are summarized in Table 8. The winglet parameters converged to a moderate leading-edge sweep ($\Lambda_{\text{winglet}} = 19.37^\circ$) and near-neutral toe-out ($\theta_{\text{winglet}} = -0.09^\circ$), which diffuses tip vortex circulation to minimize induced drag at cruise lift. The outward cant combined with moderate leading-edge sweep provides a restoring yawing moment contribution during sideslip ($\beta \neq 0$), partially compensating the weak Dutch roll damping inherent to tailless configurations without the profile drag penalty of separate vertical fins; quantification of the directional weathercock derivative $C_{n\beta}$ requires sideslip-resolved analysis and is deferred to the lateral-directional dynamics study outlined in Section 8.

8. Conclusion

This paper presented a physics-regularized, surrogate-assisted multidisciplinary design optimization methodology for long-range tailless Blended Wing Body UAVs. The framework addresses three interconnected challenges: the absence of dedicated pitch trim surfaces, severe non-linear coupling between dynamic center-of-gravity placement and reflexed aerodynamic camber, and design space contamination caused by unconstrained geometric parameterizations in low-order potential flow solvers.

The central contribution is the empirical characterization and mathematical elimination of a geometric dead zone in tailless MDO. A 1,000-sample Latin Hypercube ablation study demonstrated that unconstrained chord parameterizations allocate 35.5% of evaluations to chord-inverted configurations that achieve 29.0% lower maximum aerodynamic efficiency ($p < 0.001$)—critically, without triggering numerical solver divergence. A C^1 -continuous Hermite geometric manifold enforcing strict chord monotonicity ($c_{\text{root}} \geq c_{\text{blend}} \geq c_{\text{tip}}$) for aft-swept configurations eliminates this dead zone by construction, focusing all computational budget on the physically productive design subspace. Coupled with this manifold, an automated closed-loop mass-centroid balance solver dynamically inverts internal battery placement at every function evaluation, guaranteeing positive static margin ($SM = 22.5\%$) and near-zero cruise pitching moment ($C_m = -0.019$) at zero elevon deflection.

The framework was demonstrated across a 10,087-evaluation benchmark campaign. A 5-MLP ensemble surrogate [33] achieved 5-fold cross-validation $R^2 \geq 0.991$ across all aerodynamic and stability targets with 92.5% precision Random Forest feasibility gating; 287 active learning iterations delivered a +7.5% range improvement (+20.9 km). Cross-solver consistency analysis against viscous 3D strip theory (AeroBuildup) demonstrated that the VLM pipeline consistently underpredicts L/D relative to the higher-fidelity solver ($L/D = 17.12$ versus 23.22), behaving as a conservative estimator for attached-flow cruise drag within the employed computational hierarchy. The resulting Pareto champion achieves an operational range of 298.7 km under baseline energy assumptions and 340.7 km under verified commercial 21700 cell specifications.

8.1. Limitations and Future Work

Several limitations of the present study merit acknowledgment. The aerodynamic optimization relies on potential-flow VLM corrected with 2D profile drag strips; 3D Reynolds-Averaged Navier–Stokes (RANS) CFD validation and wind-tunnel testing are warranted to confirm detailed spanwise boundary-layer crossflow behavior and trailing-edge viscous decambering across the flight envelope. The ablation study in Section 3 compares sampling distributions under unconstrained versus manifold-constrained LHS; a controlled end-to-end optimization campaign under both parameterizations would more precisely quantify the manifold’s contribution to final Pareto quality. The present study is scoped exclusively to longitudinal stability and trim; lateral-directional dynamics—including Dutch roll damping (which is typically weakly damped in tailless configurations lacking vertical tail surfaces), spiral mode stability, and adverse yaw coupling during coordinated turns—are critical for tailless BWB aircraft with winglets and constitute a distinct analysis requiring coupled 6-DoF flight dynamics simulation. Clean-inflow propeller performance was modeled; propulsion installation effects arising from upstream center-body boundary layer wake ingestion and propeller slipstream interaction with the elevon control surfaces remain to be characterized experimentally. The present 10,087-evaluation campaign served as an exhaustive computational testbed for active learning and feasibility classifier benchmarking; transferring this architecture to high-fidelity RANS environments with restricted sample budgets ($O(10^2\text{--}10^3)$) represents a primary next step. Structural mass estimation utilized empirical component relationships calibrated to small composite UAV mass fractions (0.25–0.35) [27] rather than high-fidelity finite element analysis; the sensitivity analysis in Section 7 confirms robustness to $\pm 50\%$ airframe mass perturbations, but a detailed spar/skin composite layup analysis using FEA would provide a more rigorous structural validation. The MH91 and MH60 reflexed airfoils employed in this study originate from hobbyist-developed aerodynamic databases [31] without peer-reviewed wind tunnel characterization at the operating tip-section Reynolds numbers ($Re \approx 3.5 \times 10^5$); the geometric manifold framework is airfoil-agnostic and can accommodate any validated profile set. Dynamic stability mode damping (phugoid, short-period, lateral-directional Dutch roll with winglets) and elevon control authority under atmospheric turbulence will be evaluated through 6-DoF flight dynamics simulations and subscale prototype flight testing.

CRediT Author Statement

Mudit Atrey: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The optimization dataset comprising 10,087 aerodynamic and mission evaluations, surrogate training code, and evaluation scripts will be made available in a public repository upon acceptance. Pre-publication access is available from the corresponding author upon reasonable request.

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